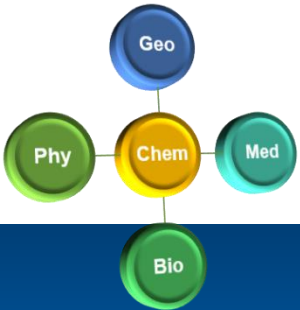


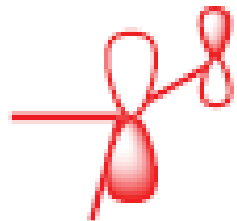


Chapter 6

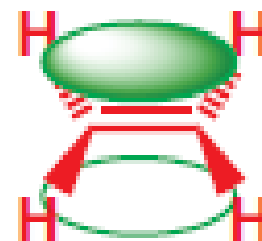
π 键上的反应 (II):

烯烃和炔烃的亲电加成反应

alkene
LUMO

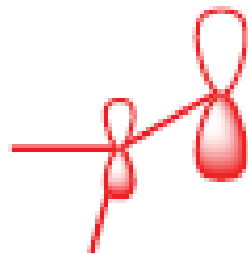


alkene = nucleophile



HOMO = filled π orbital

alkene
HOMO





本章内容

6.1 烯烃与溴单质的反应

6.2 烯烃的环氧化反应

6.3 非对称烯烃亲电加成反应的选择性

6.4 二烯的亲电加成反应

6.5 非对称溴鎓离子开环反应的选择性

6.6 立体专一的烯烃亲电加成反应

6.7 烯烃的双羟化反应

6.8 双键的切断反应: 高价碘氧化和臭氧化

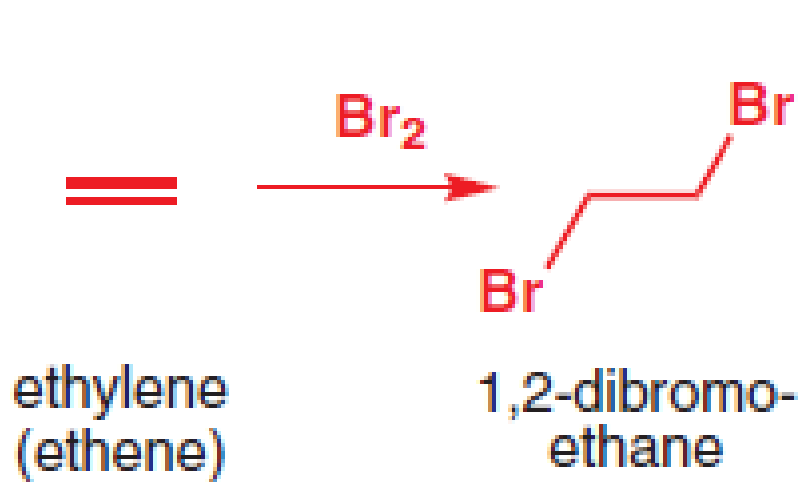
6.9 烯烃的水合反应

6.10 炔烃的亲电加成反应

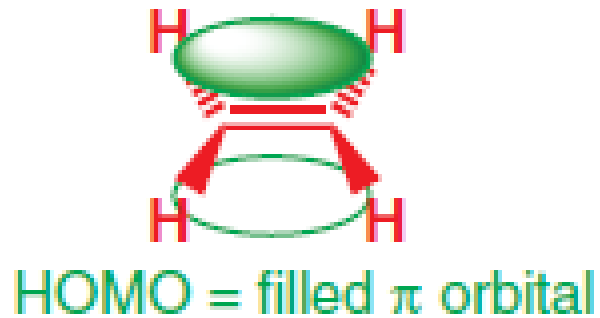
6.11 总结 **References: Chapters 19 of Clayden's Organic Chemistry**



6.1 烯烃与溴单质的反应

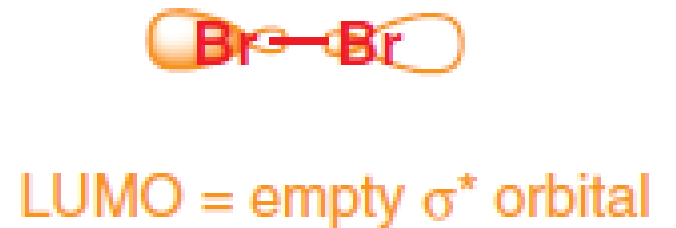


亲核试剂
alkene = nucleophile



烯烃提供HOMO

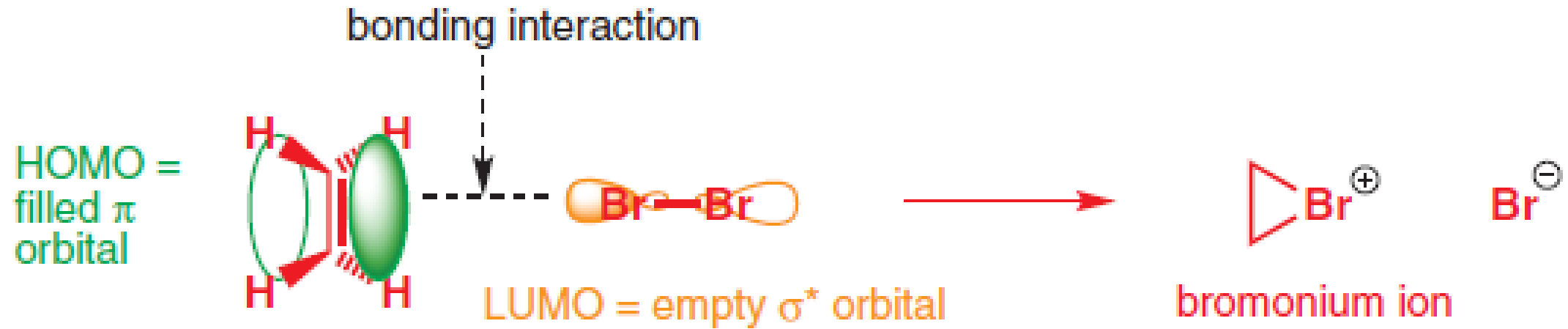
亲电试剂
 Br_2 = electrophile



溴分子提供LUMO

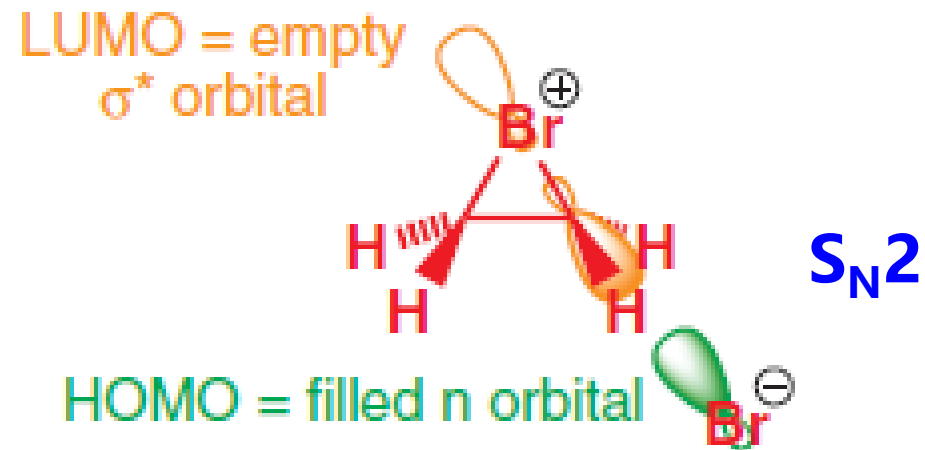
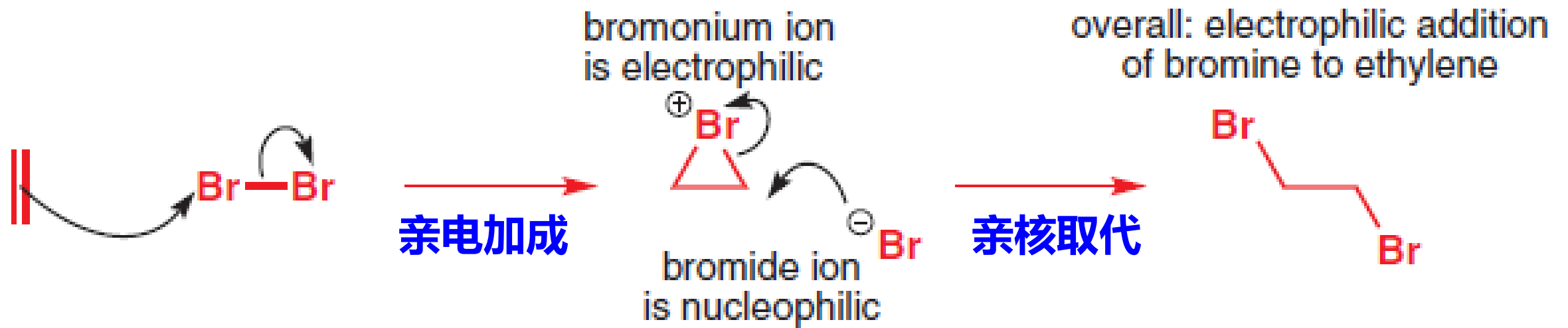


烯烃溴加成反应的机理



HOMO和LUMO相互作用

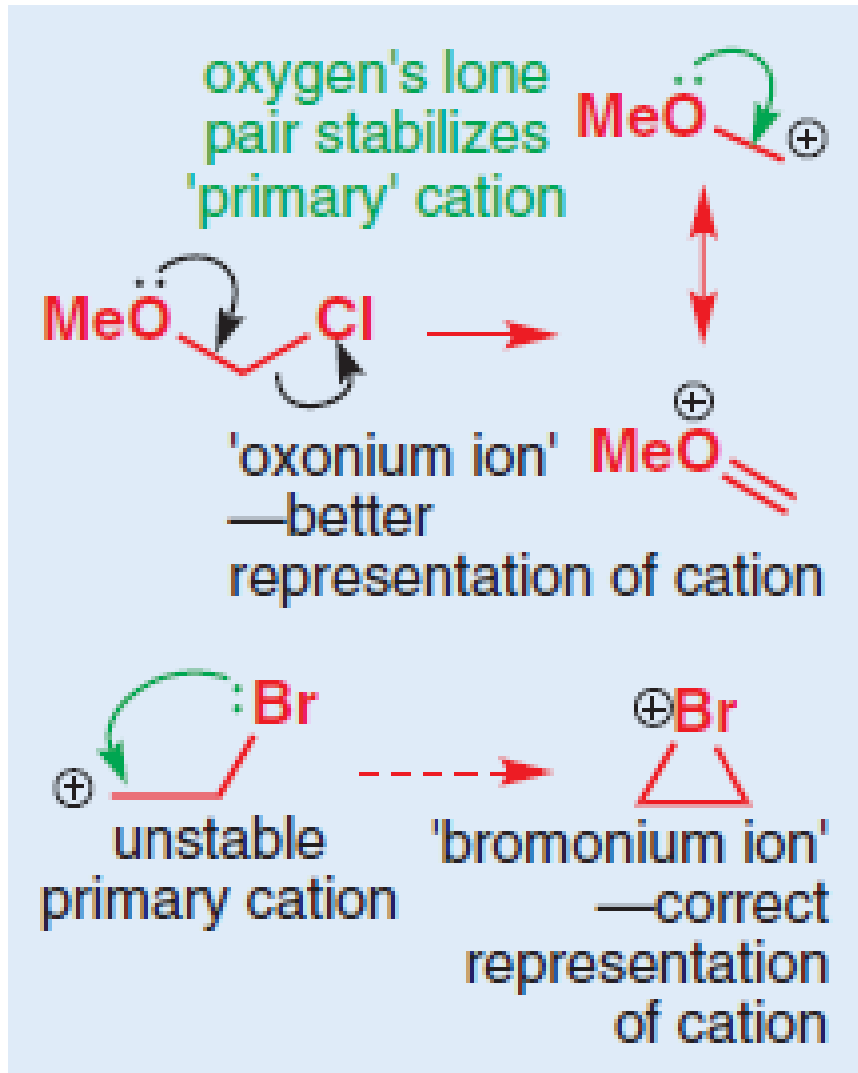




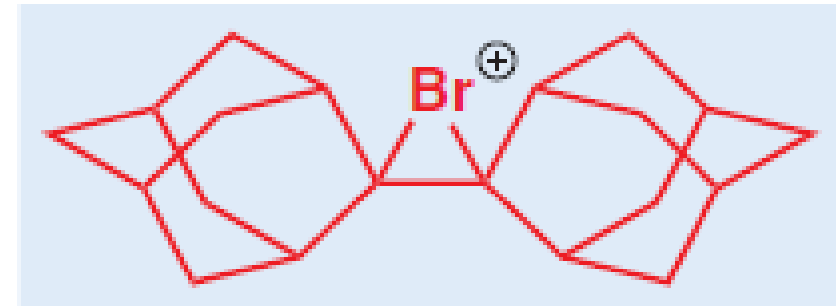


溴鎗离子

MOMCl



通常氯代物的亲核取代活性极低
而MOMCl的活性却很高

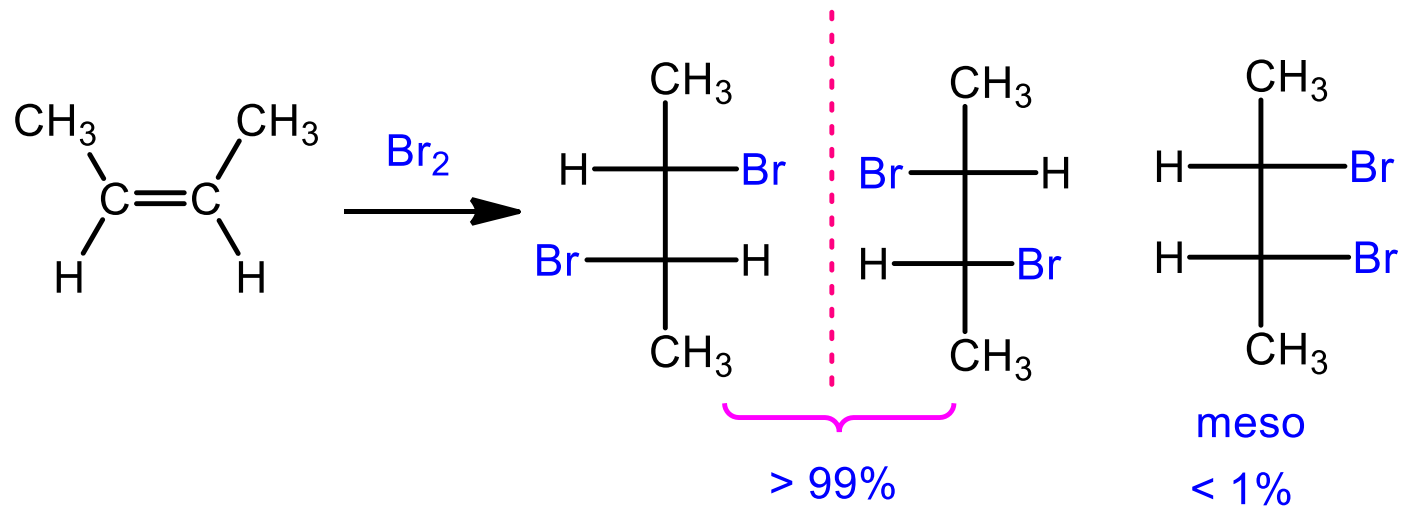
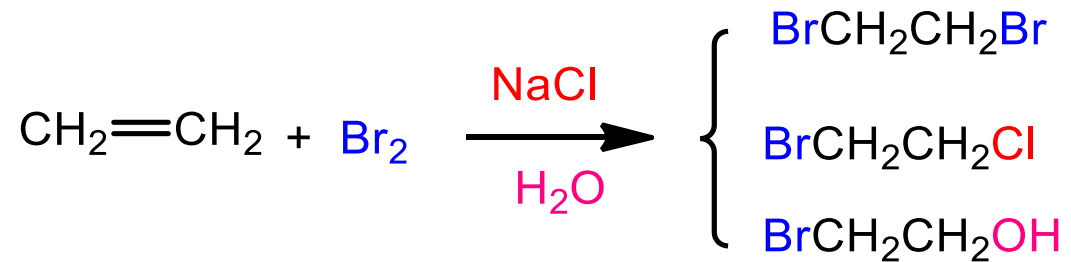
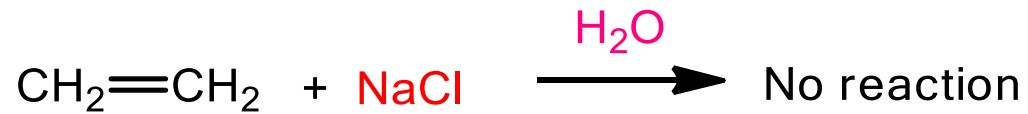


溴鎗离子相对更稳定

稳定的溴鎗离子

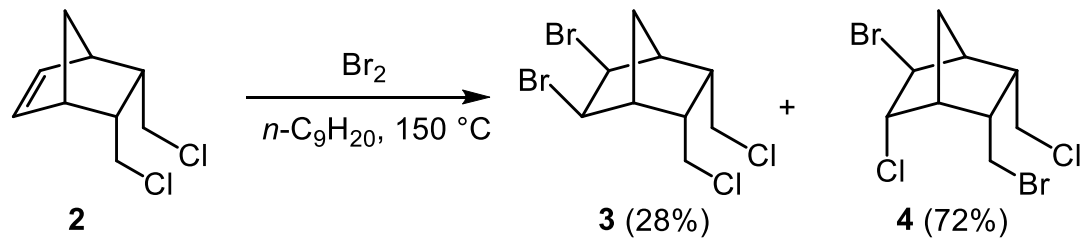


存在溴鎓离子中间体的证据

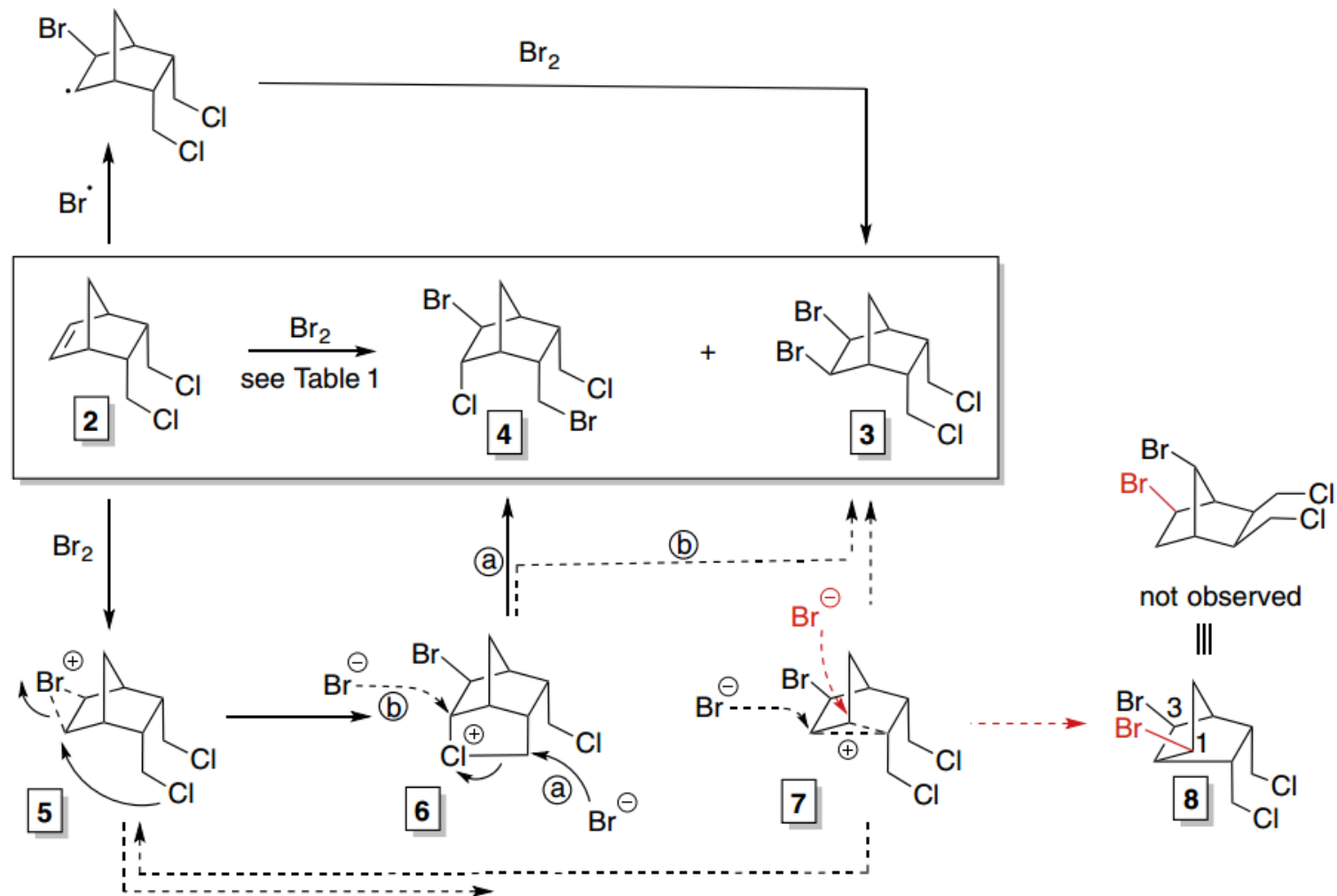




练习：画出反应机理



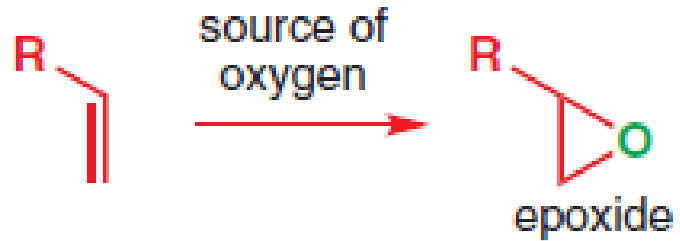
Answer:



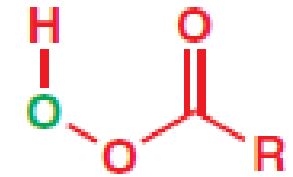


6.2 烯烃的环氧化反应

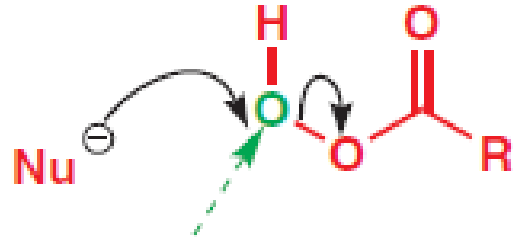
环氧化提供哪个氧?



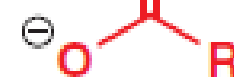
commonest oxidants in this reaction are peroxy-carboxylic acids:



过氧酸

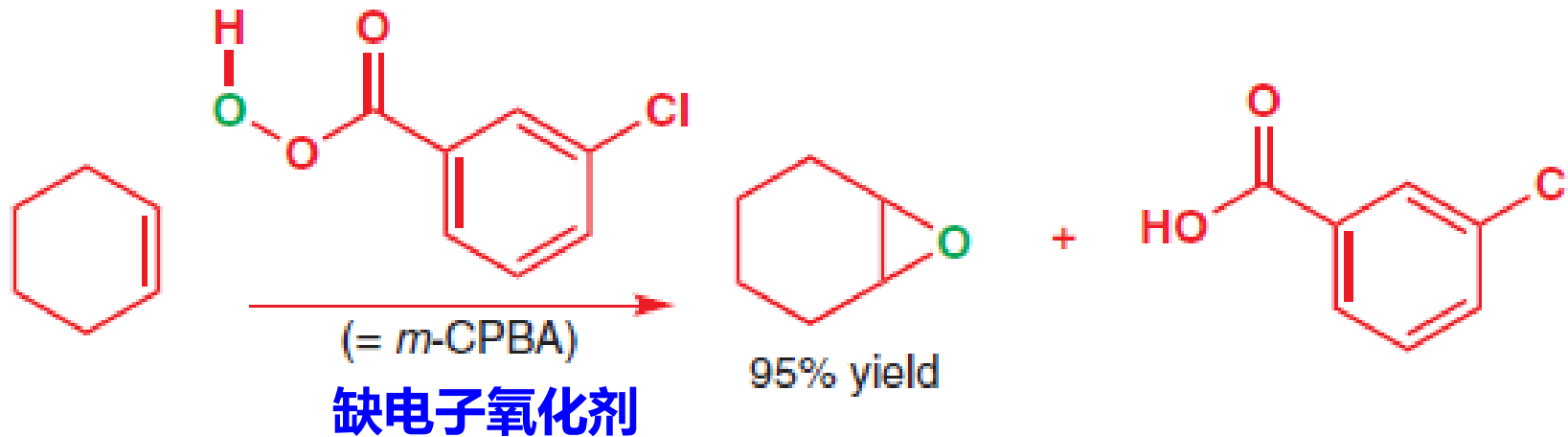


+



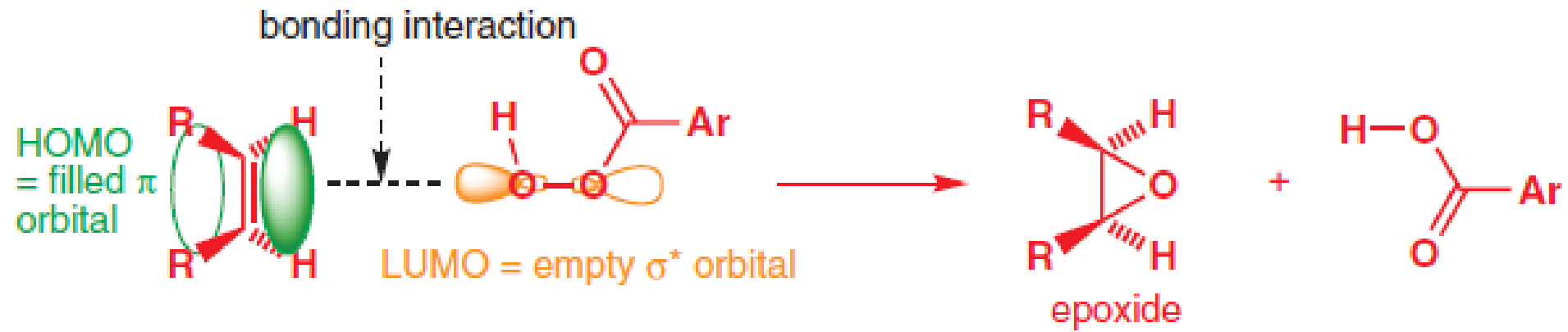
carboxylate: good leaving group

具有亲电性：氧化性（拿走电子）

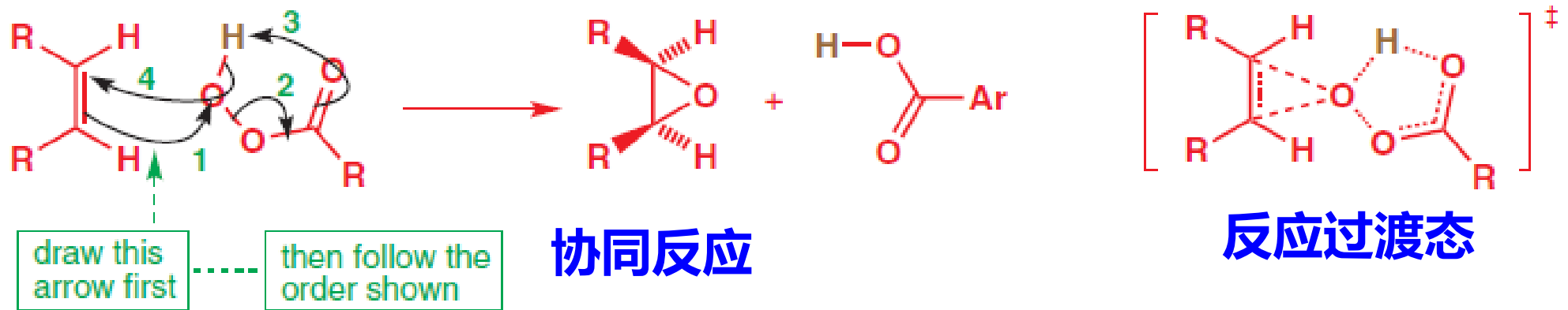




烯烃环氧化反应的机理

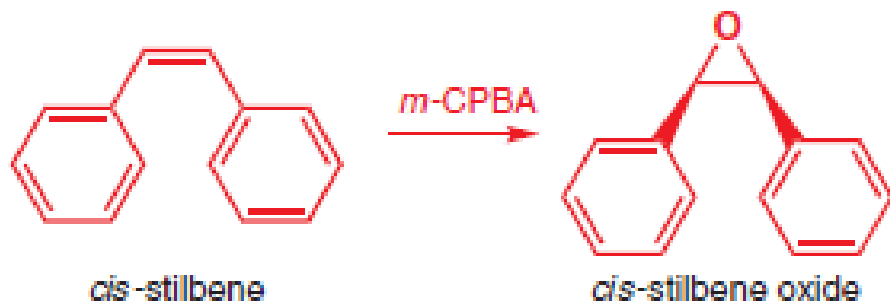


HOMO和LUMO相互作用

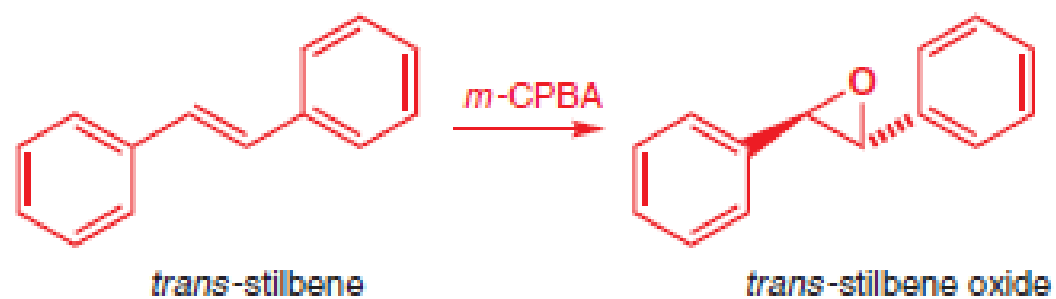




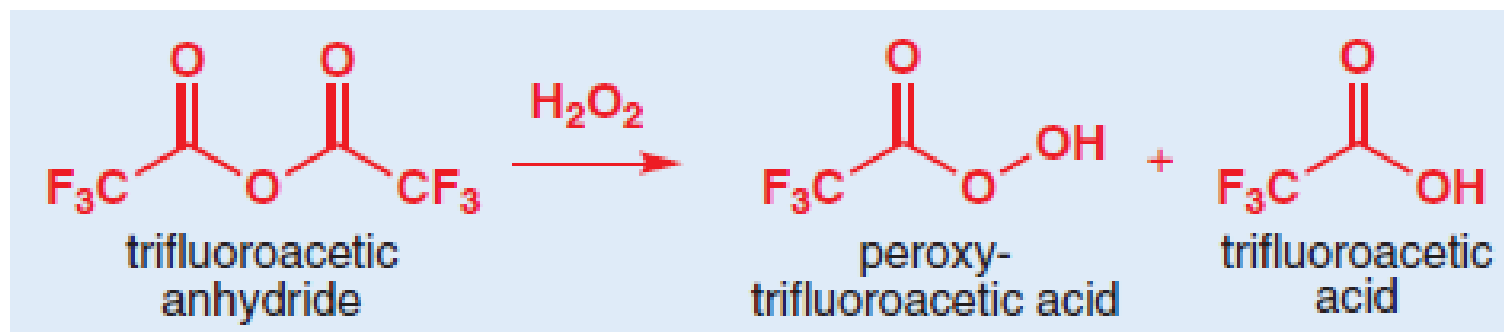
环氧化反应的立体专一性



协同反应的特征：构型保持

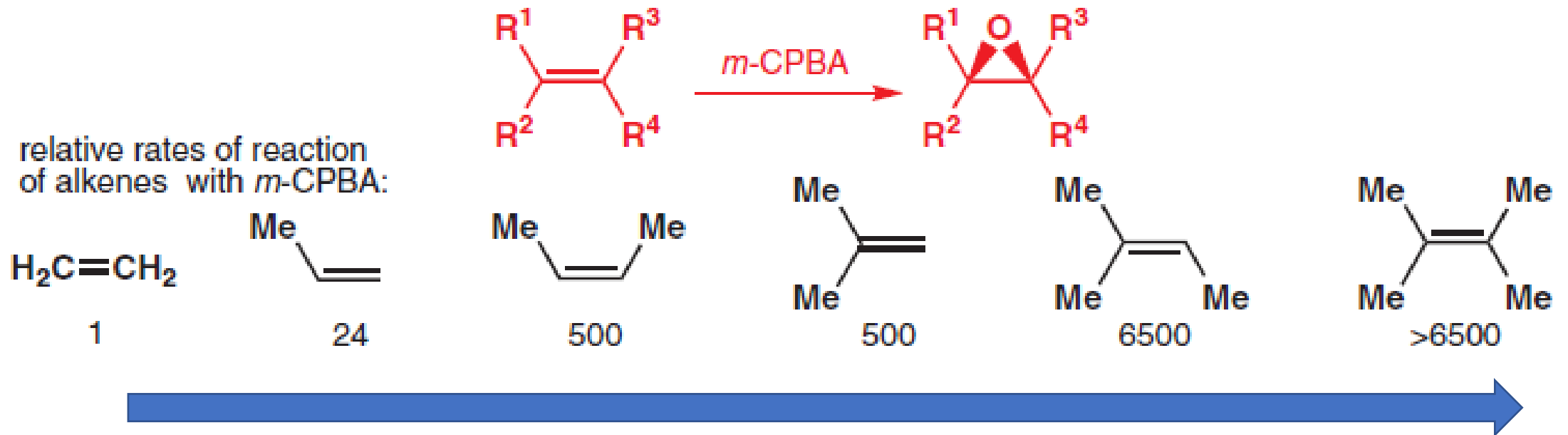


过酸的制备方法





烯烃的环氧化活性比较



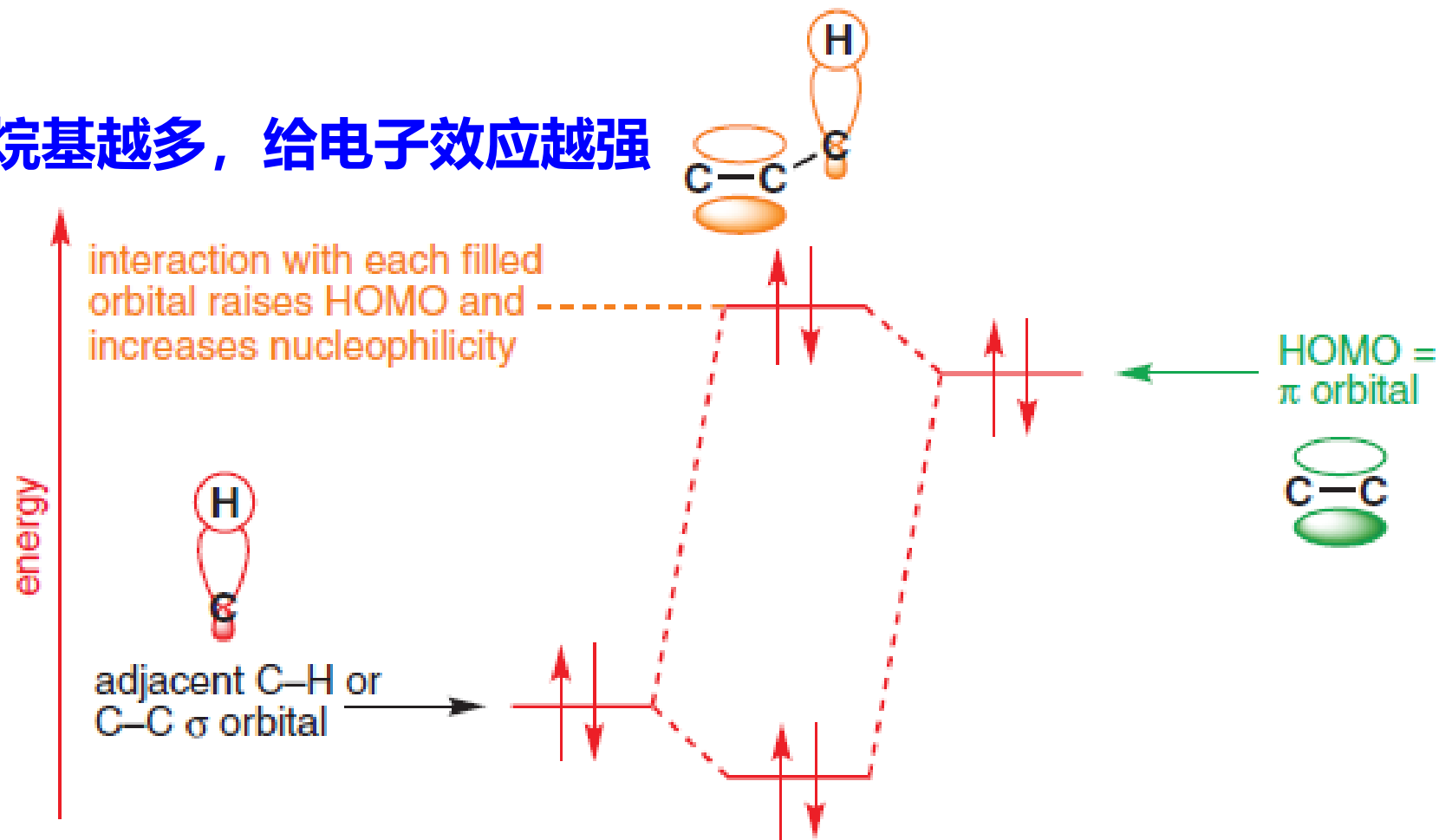
环氧化反应速率（亲电加成反应速率）

σ - π 超共轭效应：烷基对双键给电子效应的根源



σ - π 超共轭给电子效应的分子轨道解释

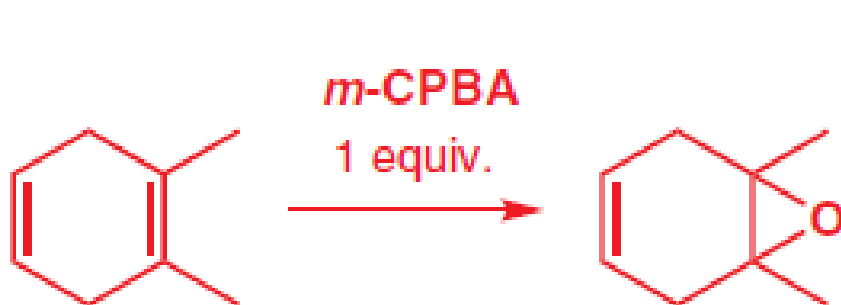
烷基越多，给电子效应越强



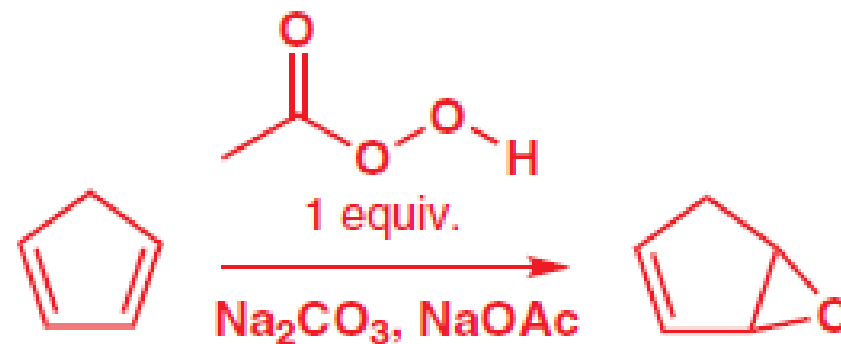
烷基- π 体系，给电子作用，提升HOMO能量，增强烯烃亲电性



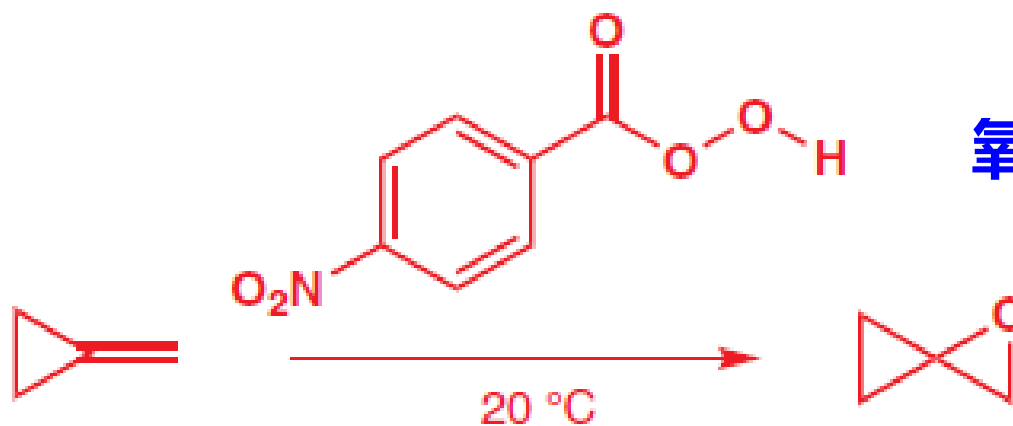
烯烃环氧化反应的选择性



环氧化反应可选择性
在更富电子双键发生



共轭二烯可选择
性实现单环氧化

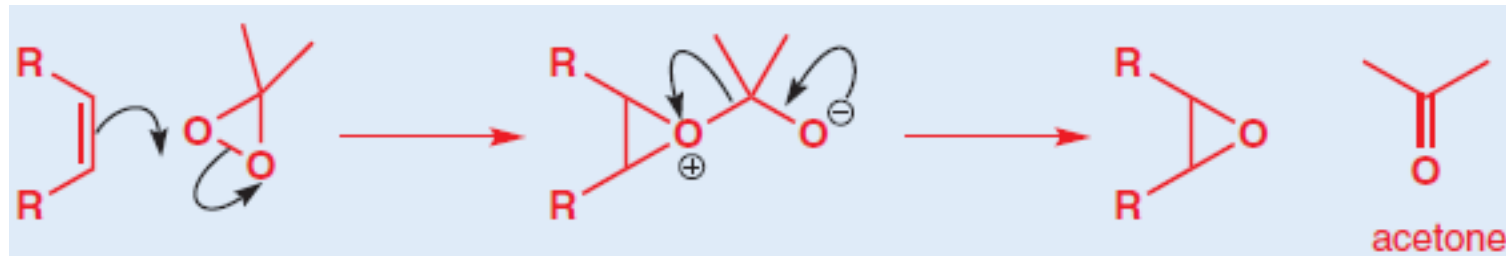
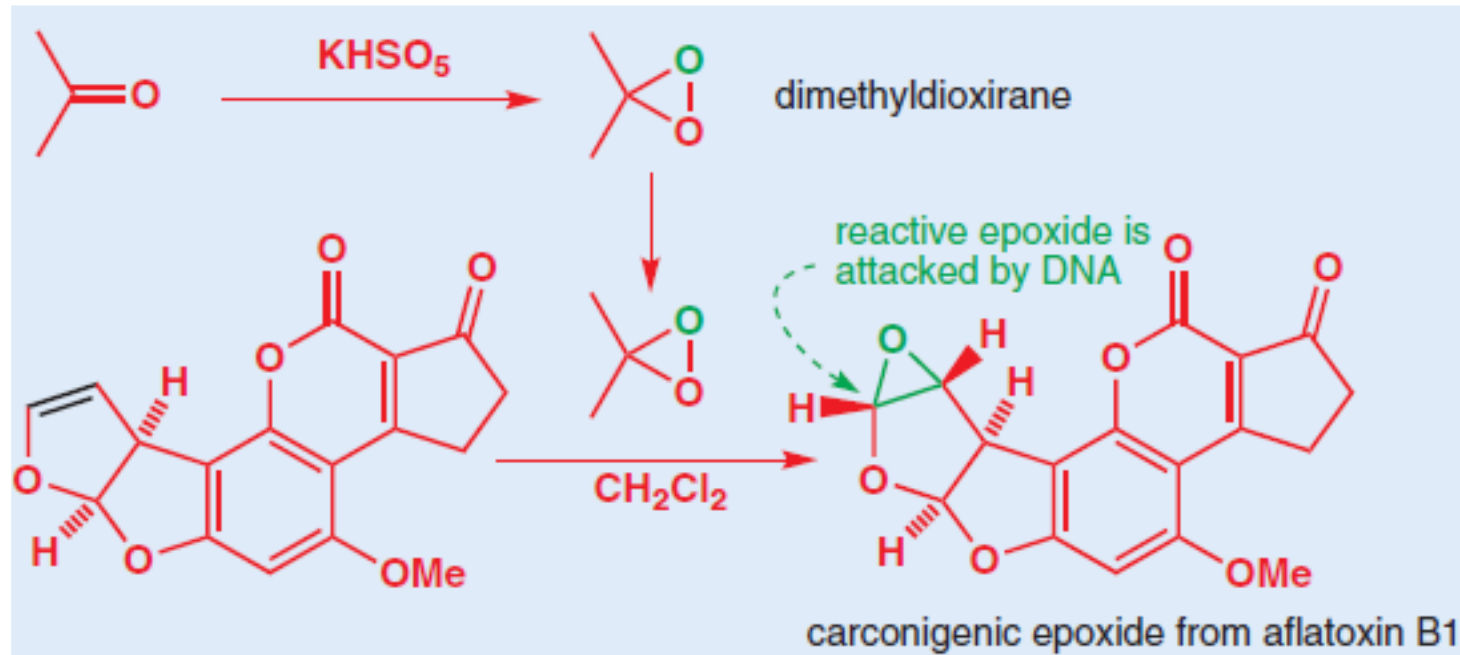


氧化性如何?

螺环环氧化物



二甲基过氧化酮 (DMDO)

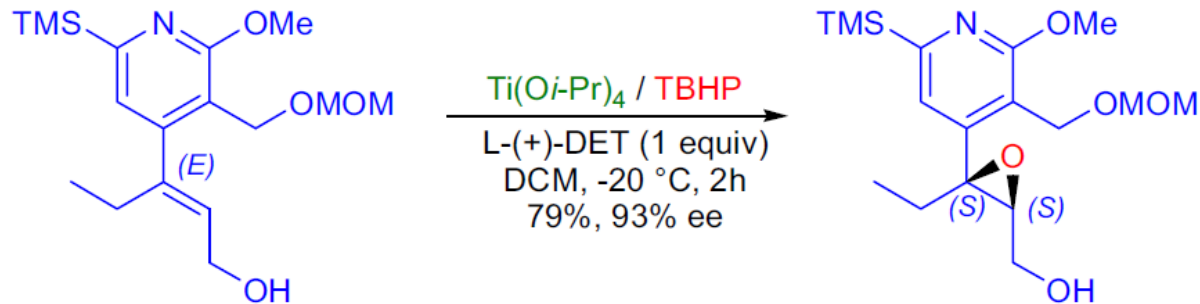


协同环氧化过程，构型保持

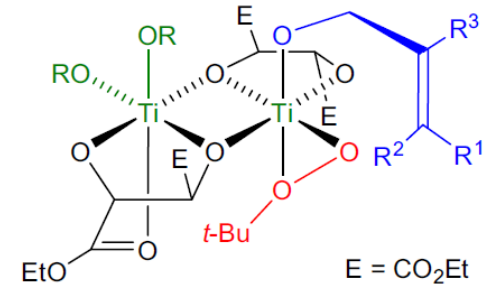


不对称环氧化 (了解性知识)

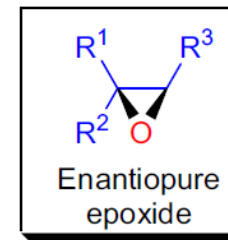
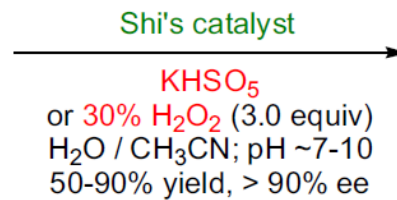
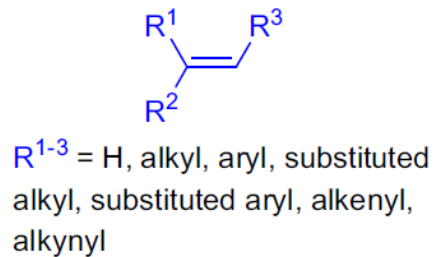
Sharpless 不对称环氧化



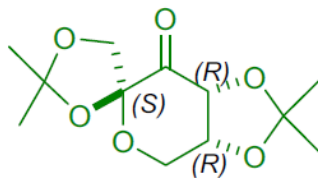
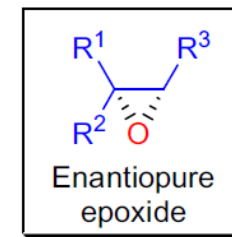
Transition state of epoxidation:



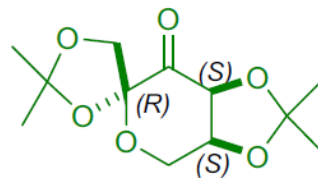
史一安不对称环氧化



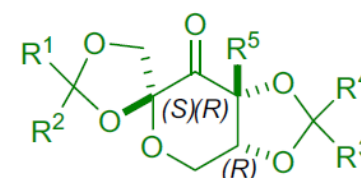
or



Shi's catalyst derived from D-fructose



Shi's catalyst derived from L-fructose



Shi's generalized catalysts (derived from D-fructose)

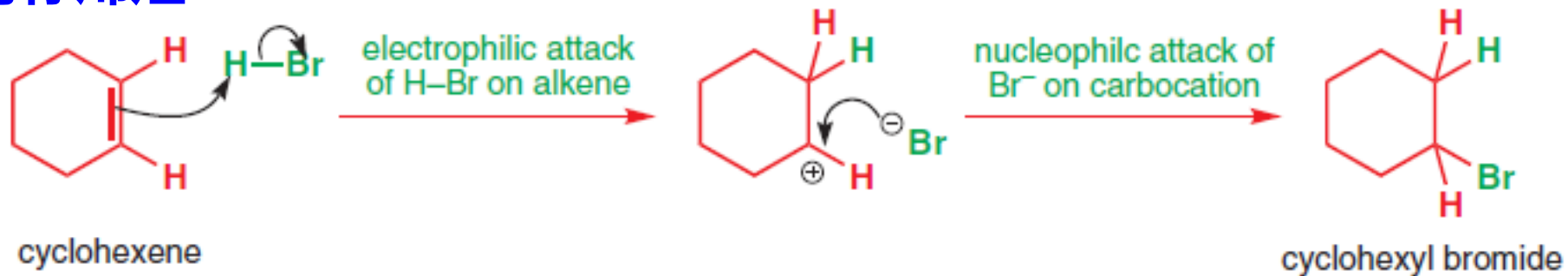
$\text{R}^{1-2} = \text{Me, Et, }-(\text{CH}_2)_4-,$
 $-(\text{CH}_2)_5-, -(\text{CH}_2)_6-$
 $\text{R}^{3-4} = \text{Me, Et, }i\text{-Pr, H, F, Bn, }-(\text{CH}_2)_4-, -(\text{CH}_2)_5-,$
 $-(\text{CH}_2)_6-$
 $\text{R}^5 = \text{H, F}$



6.3 非对称烯烃亲电加成的位置选择性

不会形成“氢鎓离子”中间体

对称烯烃

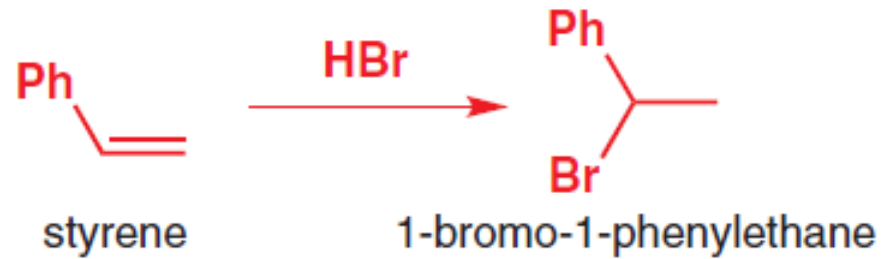


对称烯烃不存在位置选择性问题



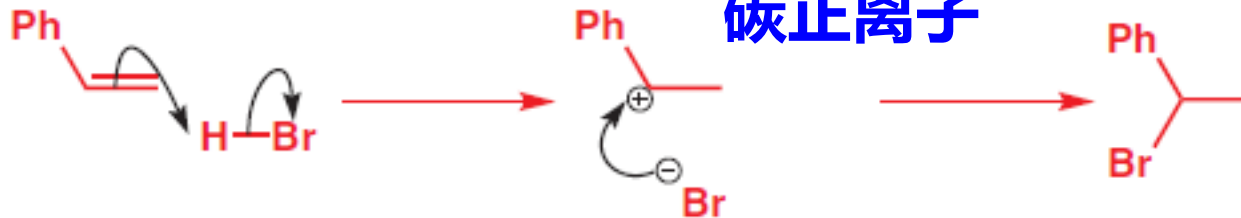
碳正离子的稳定性与亲电加成的位置选择性

非对称烯烃



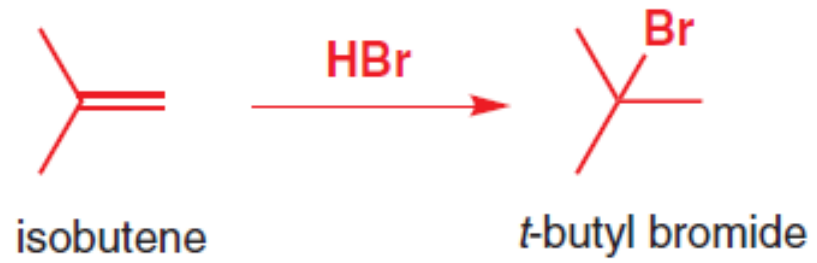
优秀的位置选择性
?

稳定的苯甲位
碳正离子

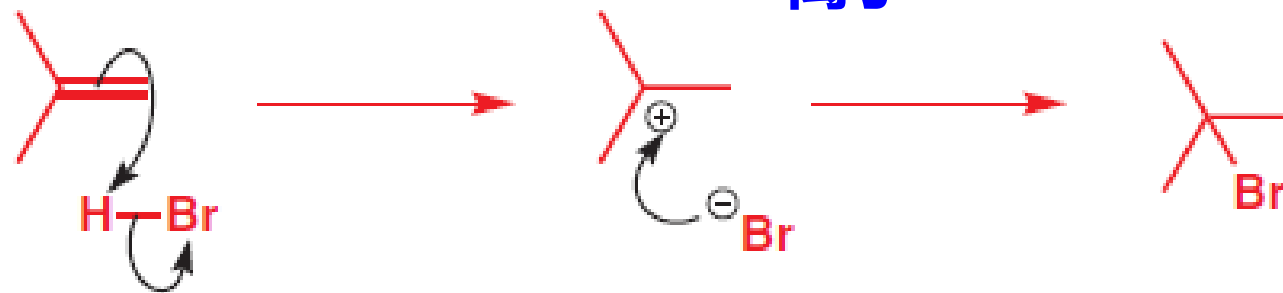


不稳定的一
级碳正离子



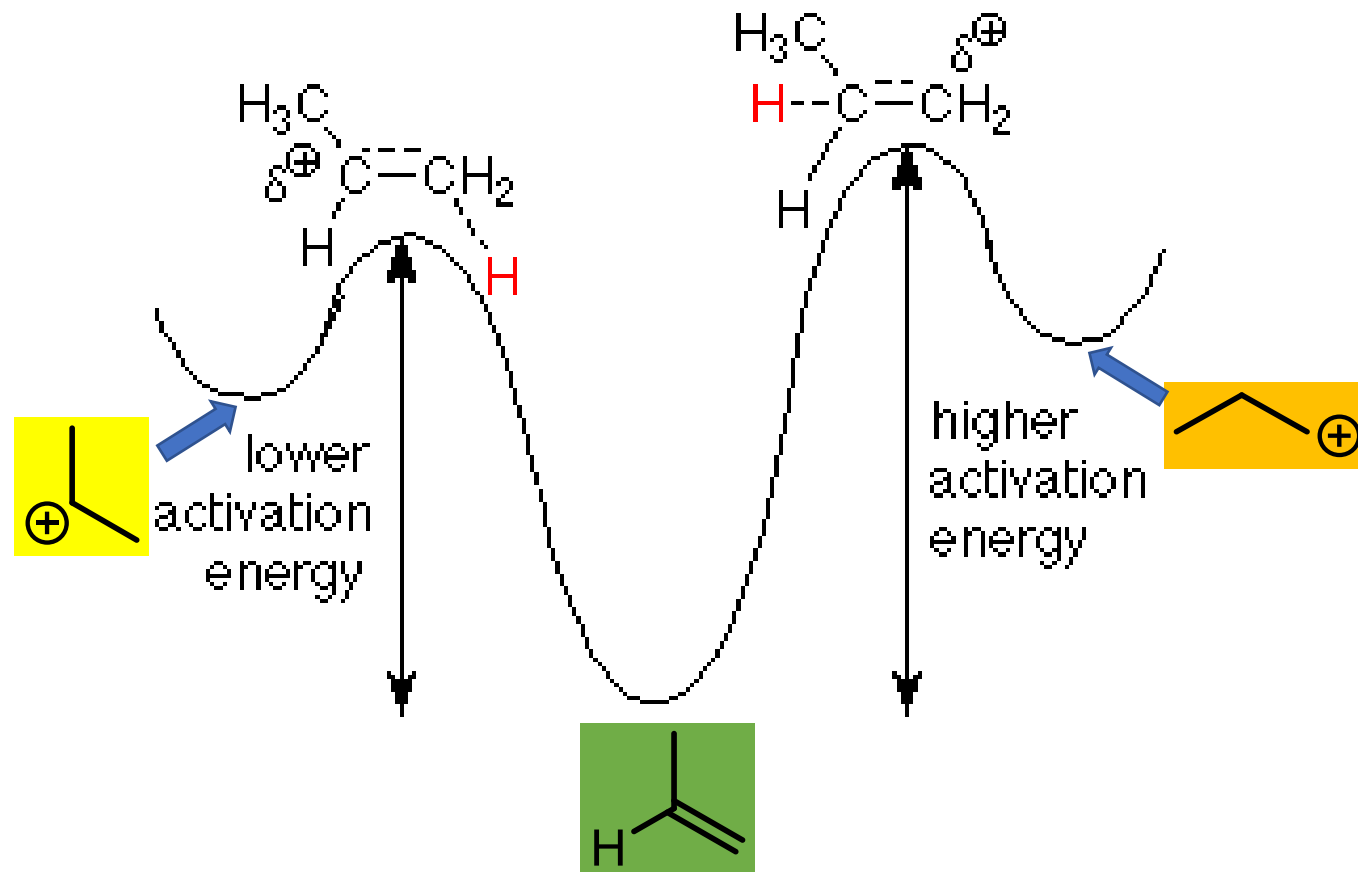
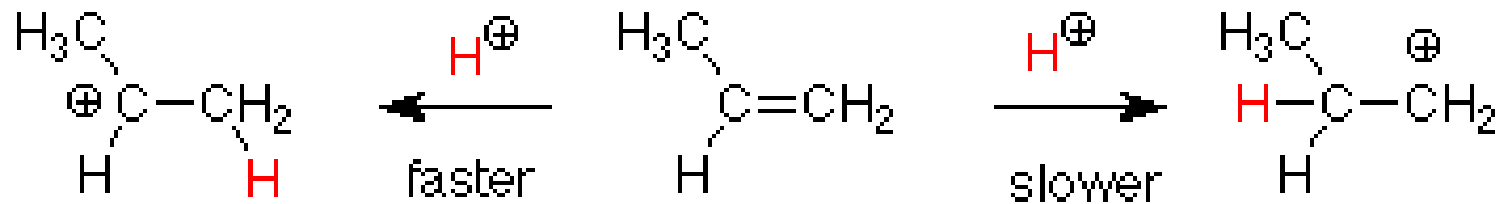


稳定的叔碳正
离子





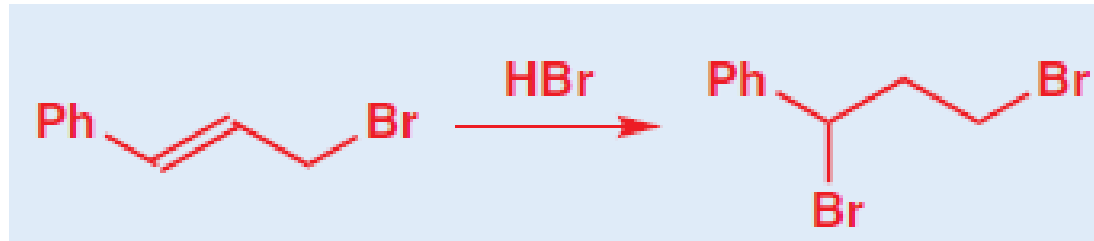
动力学因素对位置选择性的影响



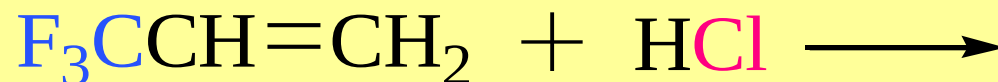
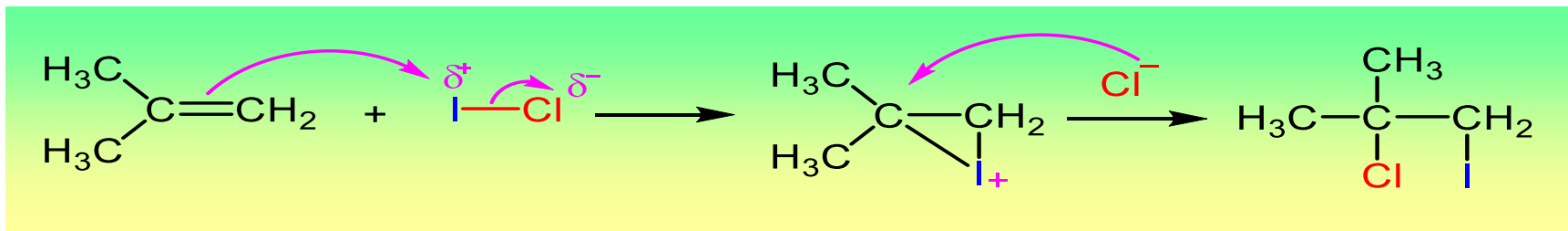


马氏规则 (Markovnikov's rule)

马氏规则: 卤化氢等极性试剂与非对称烯烃发生亲电加成反应时，酸中的**氢原子**加在**含氢较多的双键碳原子上**，其他原子（此处为卤原子）及基团加在含氢较少的双键碳原子上。



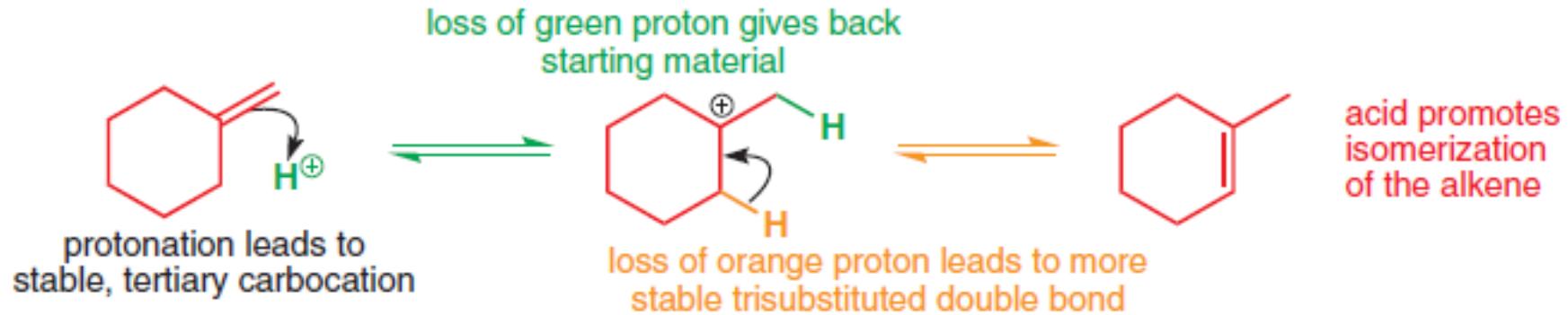
广义的马氏规则: 当X-Y与非对称烯烃发生亲电加成反应时，形成更稳定碳正离子中间体的反应途径更有利。





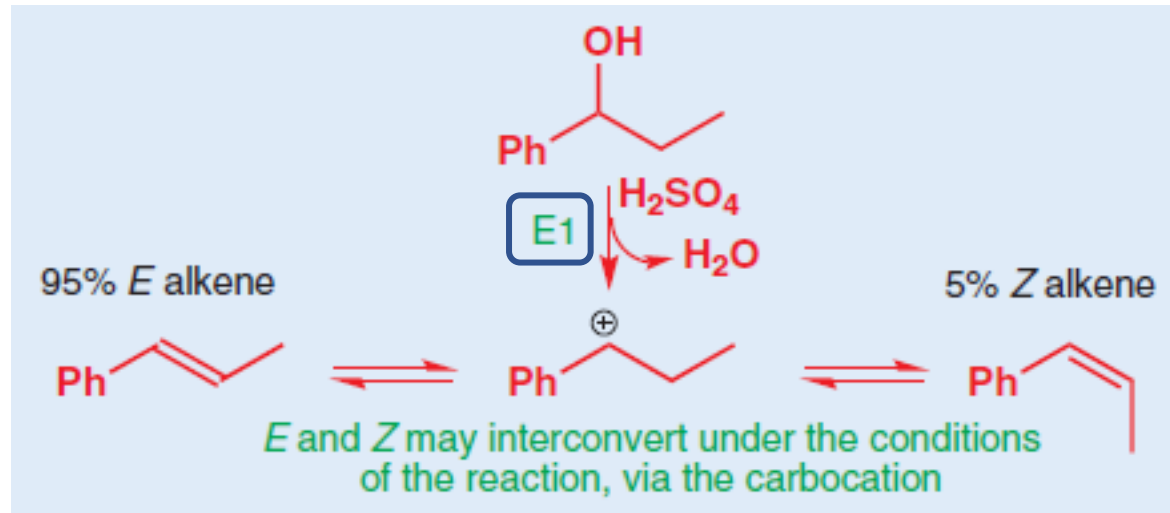
酸催化的烯烃异构反应

位置异构体



双键移位

E1消除反应

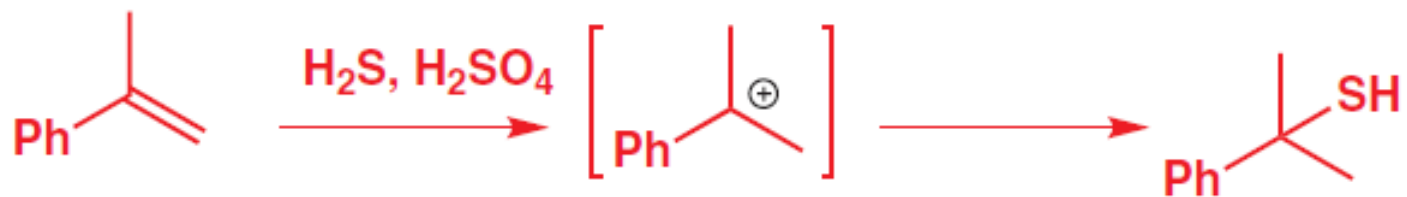
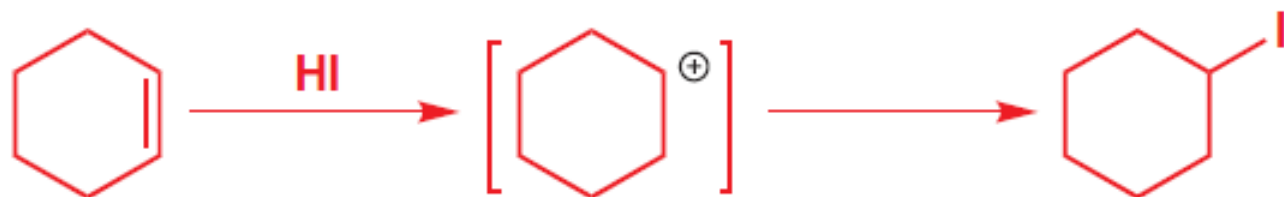


双键顺反异构

Z、E 几何异构体 (geometrical isomers)

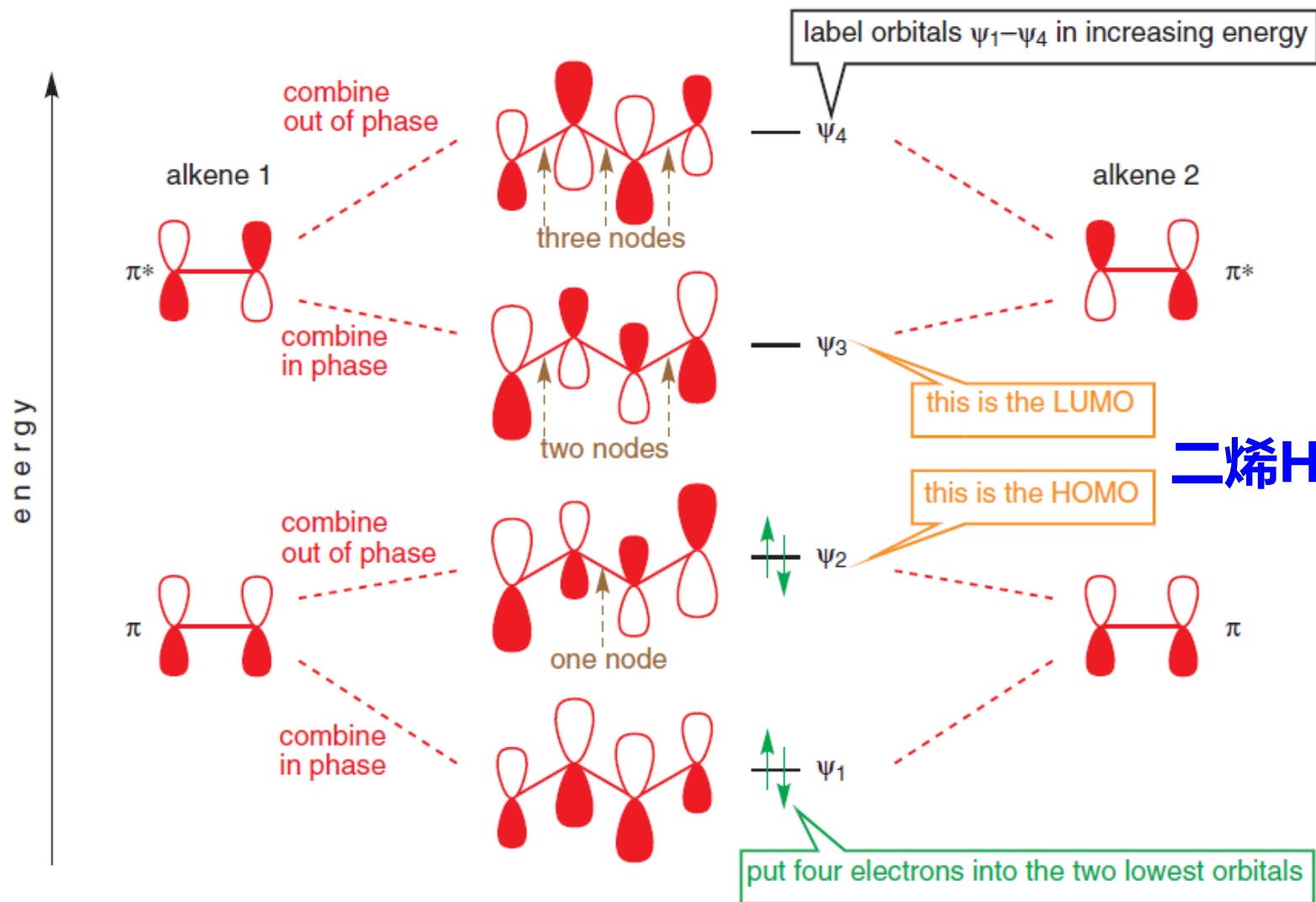


其他亲核试剂对碳正离子的捕获





6.4 二烯的亲电加成反应

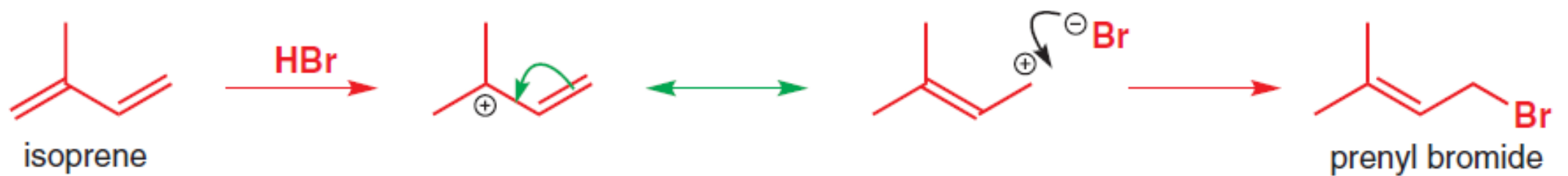
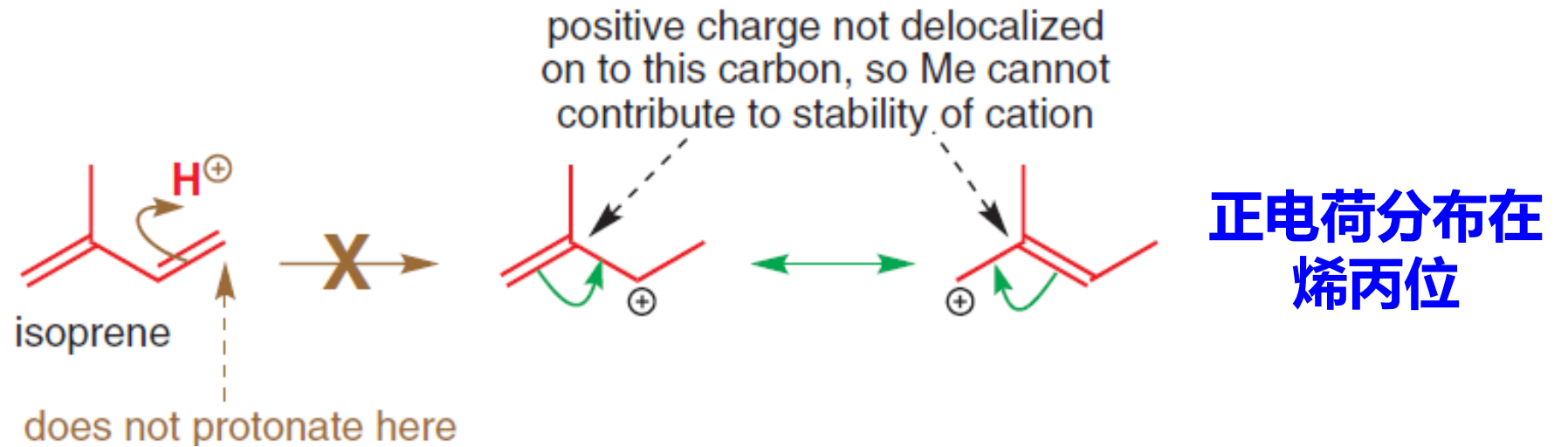
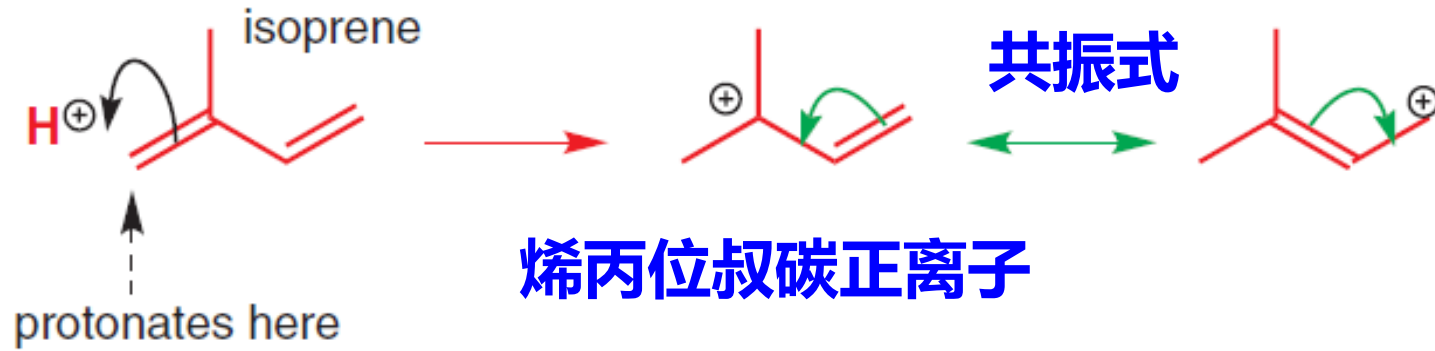


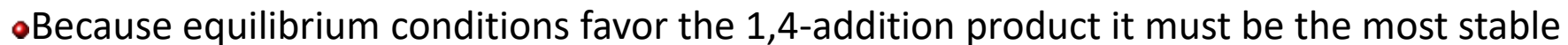
二烯HOMO能量上升
更亲电



异戊二烯的亲电加成反应

质子化的位点
选择性:







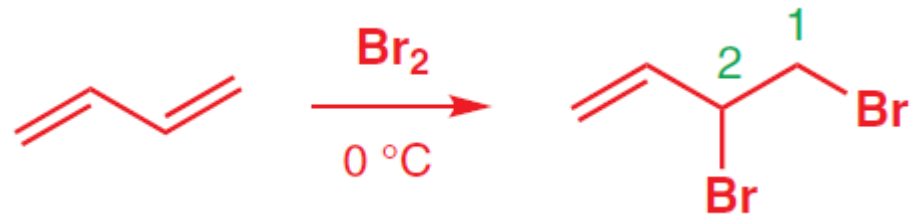
二烯的二溴加成反应

高温下



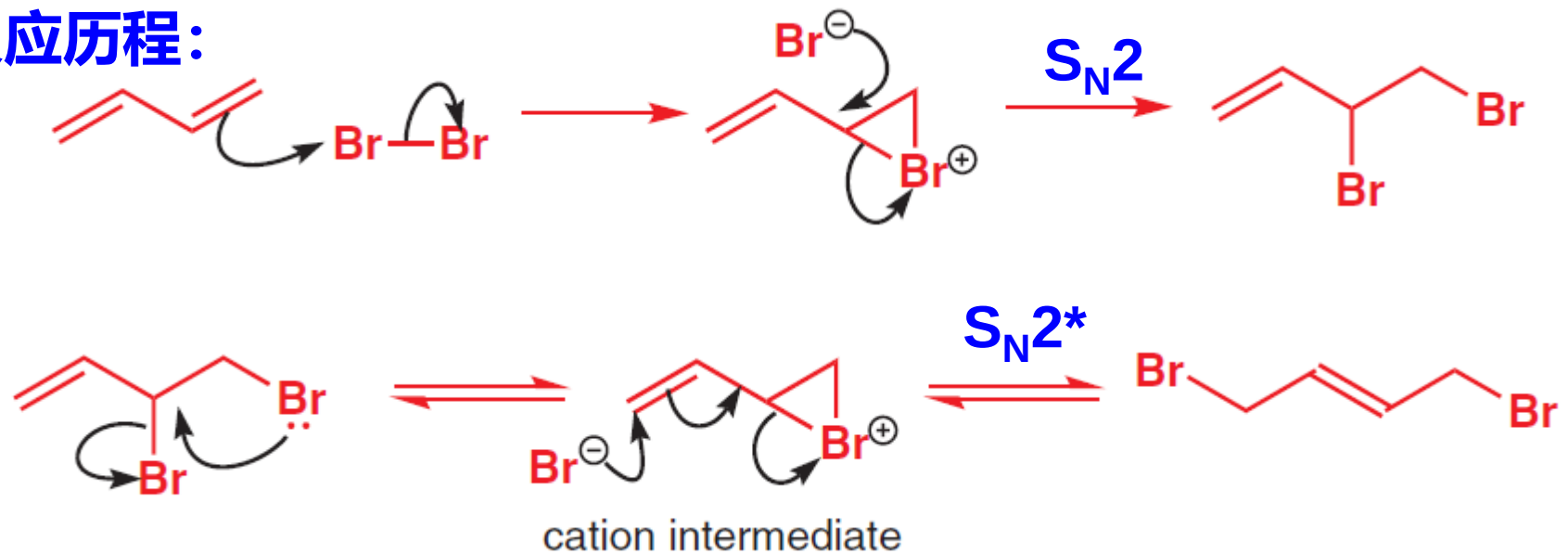
热力学产物

低温下



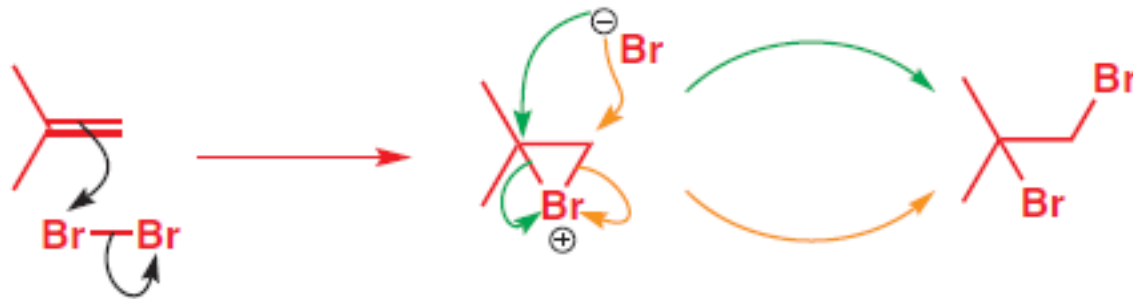
动力学产物

反应历程:

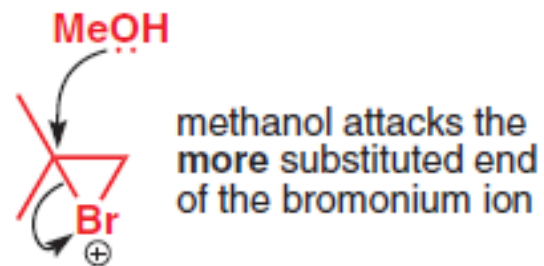
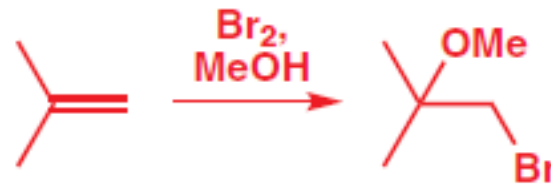




6.5 非对称溴鎓例子的位置选择性开环



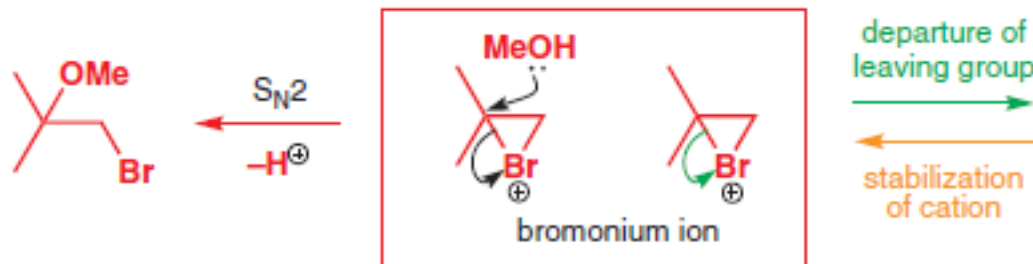
进攻两端产物相同



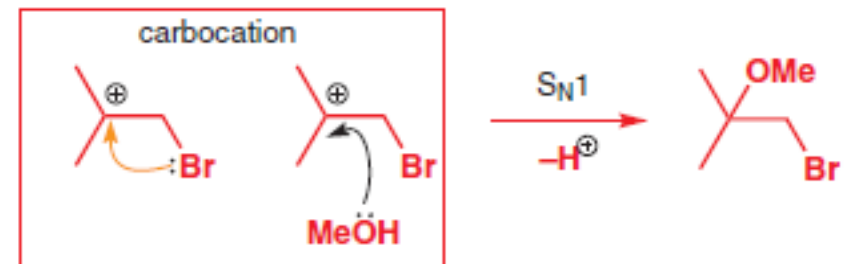
甲醇进攻溴鎓例子位阻较大一侧

两种可能机制：

two limiting mechanisms for substitution on bromonium ion



S_N2 进攻大位阻位点不正常

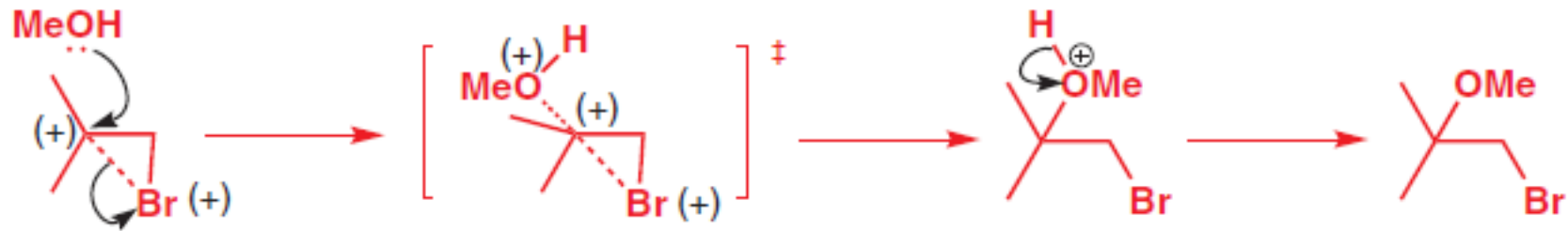
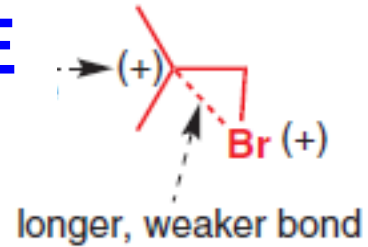


碳正离子相较于溴鎓例子不稳定



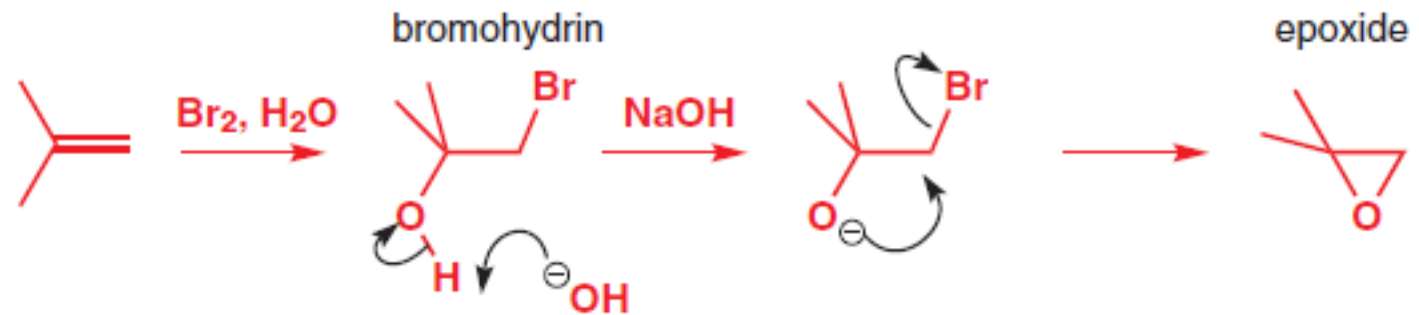
合理解释：松散的 S_N2 过渡态

部分碳正
离子



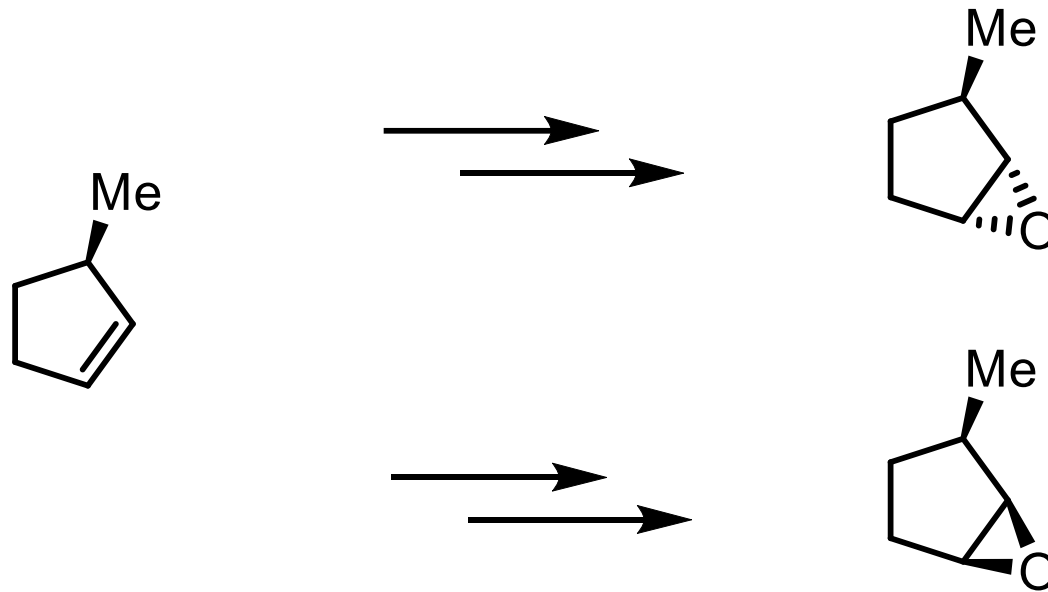
松散的 S_N2 过渡态

羟溴化：
制备环氧的另一种方法





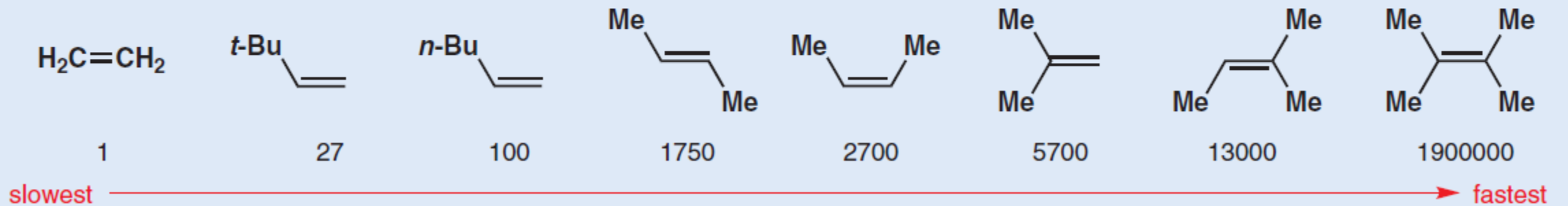
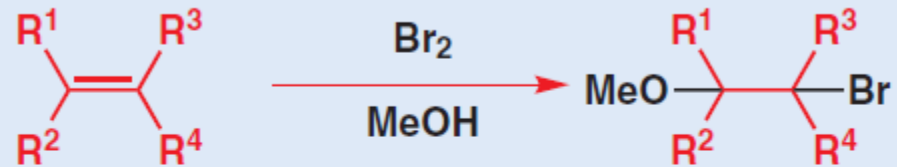
练习：完成下列转化





烯烃溴烷氧化反应的速率比较

relative rates of reaction of alkenes with bromine in methanol



从左向右，反应速率更快



环氧开环反应的位置选择性：酸性和碱性条件

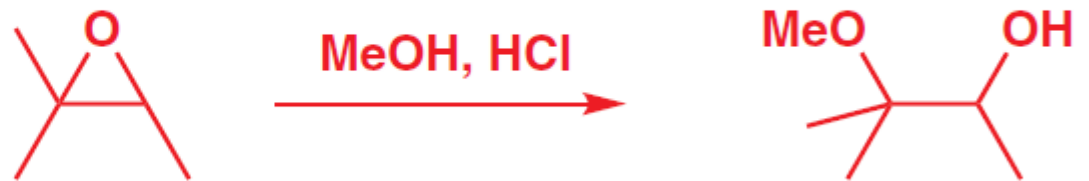
酸性和碱性条件产物不同：位置选择性不同

reaction of epoxide with
basic methoxide



attack at less substituted end

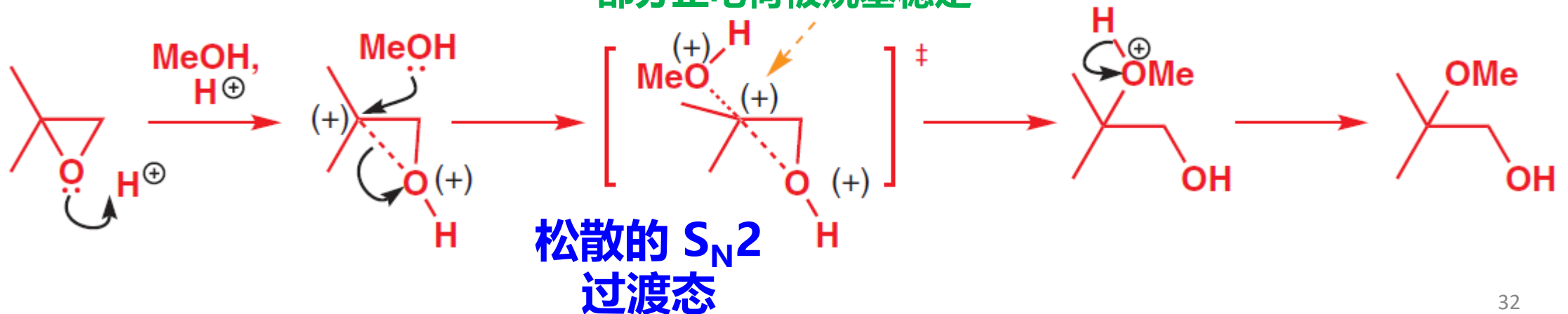
reaction of epoxide with
acidic methanol



attack at more substituted end

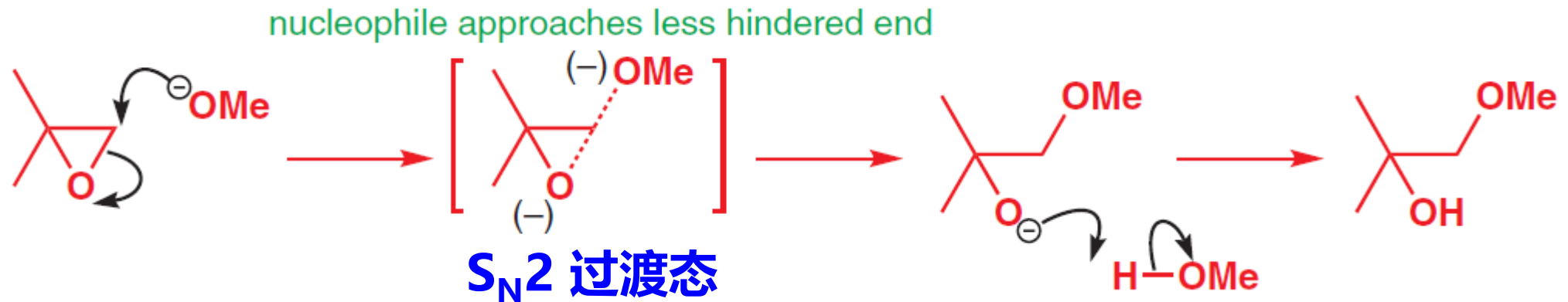
酸性条件下选择位阻较大一侧开环氧

部分正电荷被烷基稳定

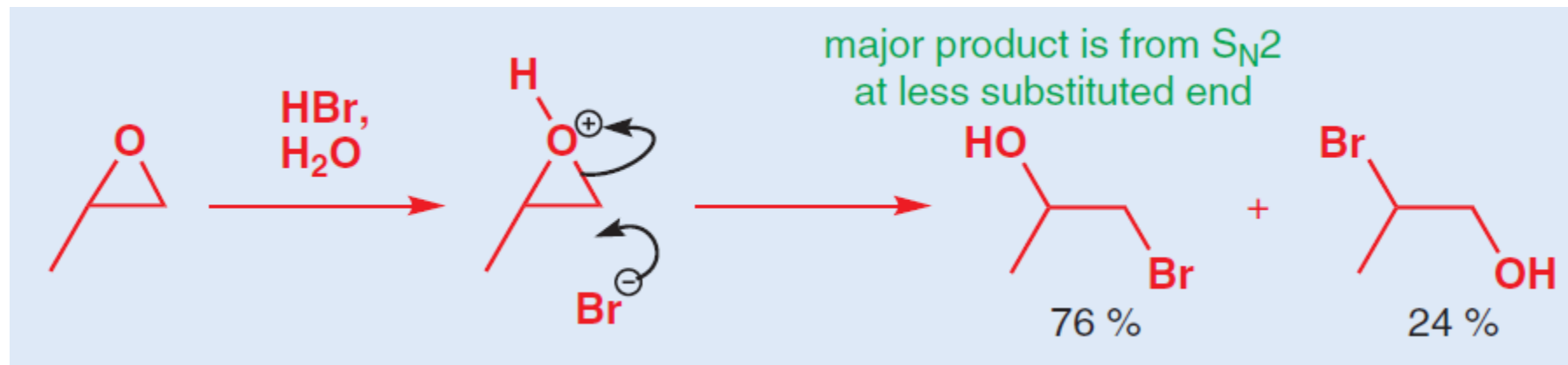




碱性条件下选择位阻较小一侧开环氧



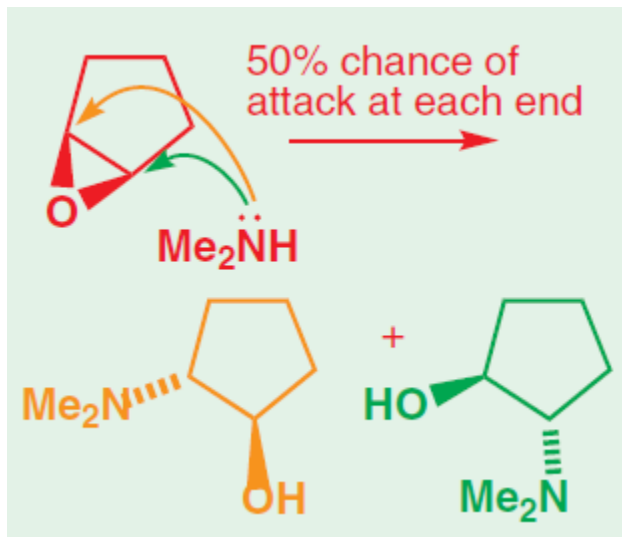
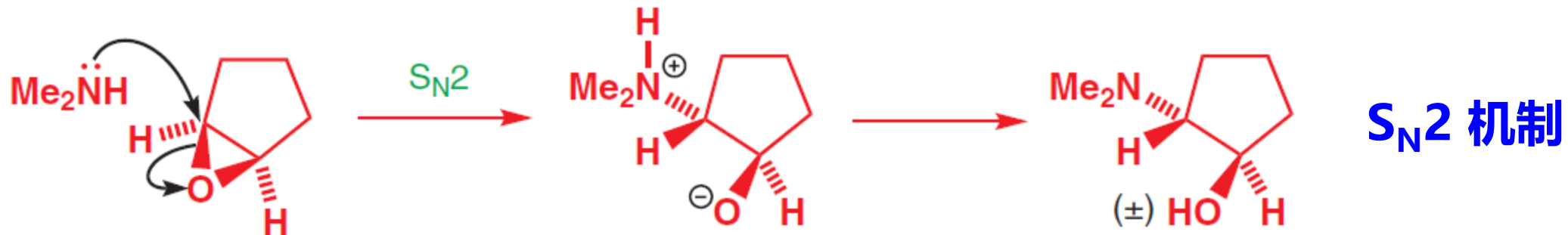
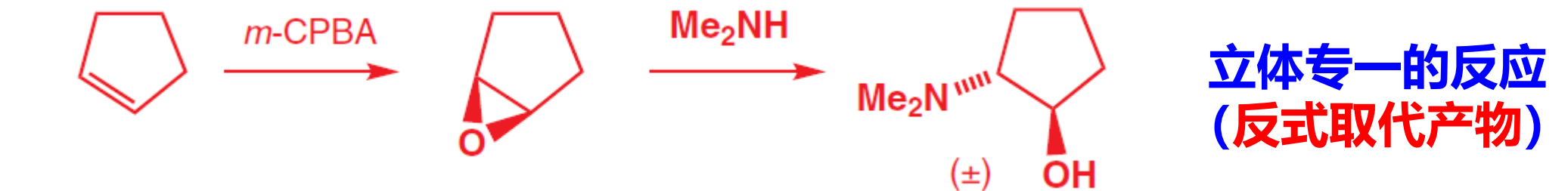
边界案例：两种产物都有



类二级碳正离子 VS S_N2



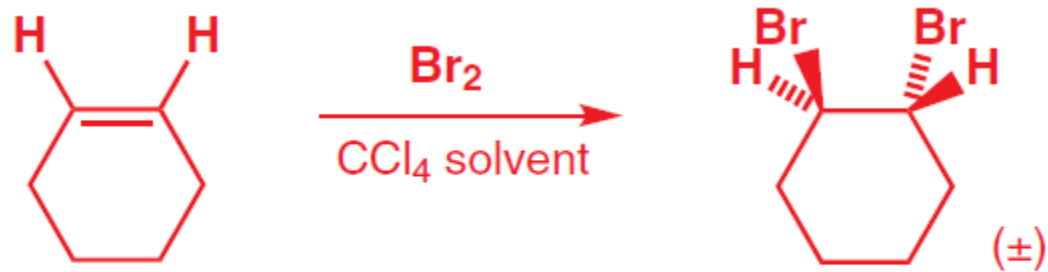
6.6 立体专一的亲电加成反应



无论进攻哪一端都只产生**反式取代产物**
(一对对映异构体)

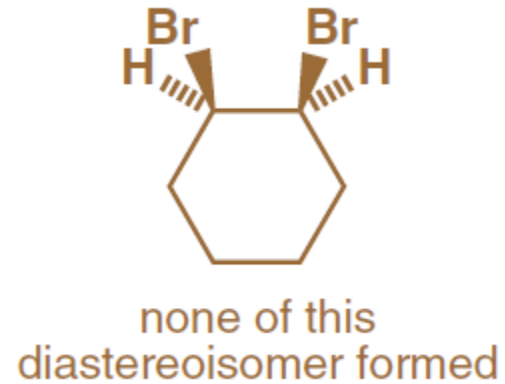


溴对烯烃的反式加成

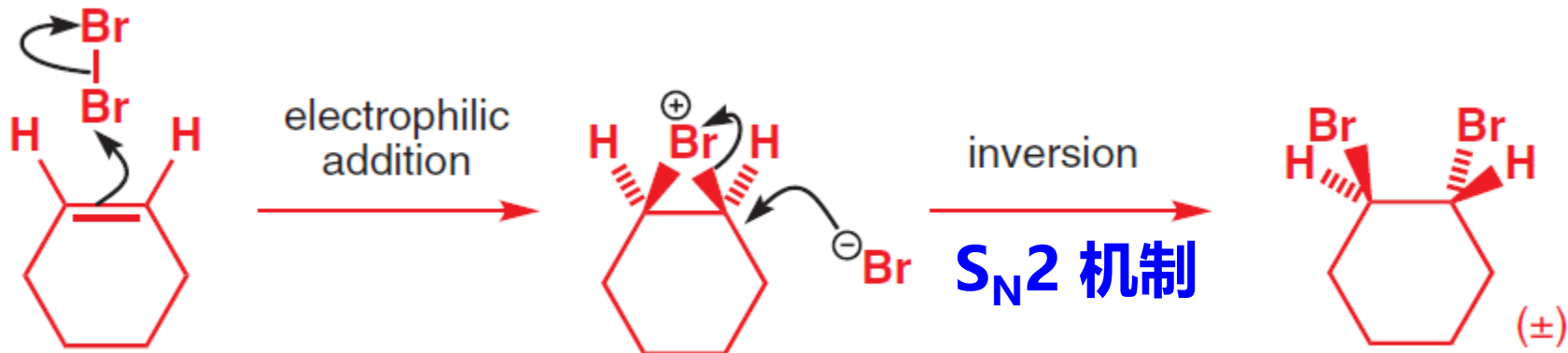


exclusively this
diastereoisomer formed

专一产生反式二溴产物



未发现顺式二溴产物

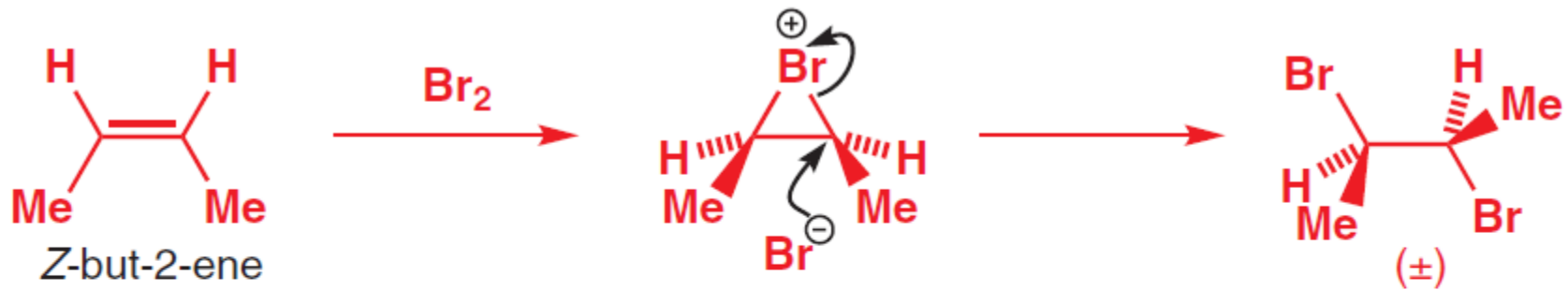


溴鎓离子中间体



溴的反式加成

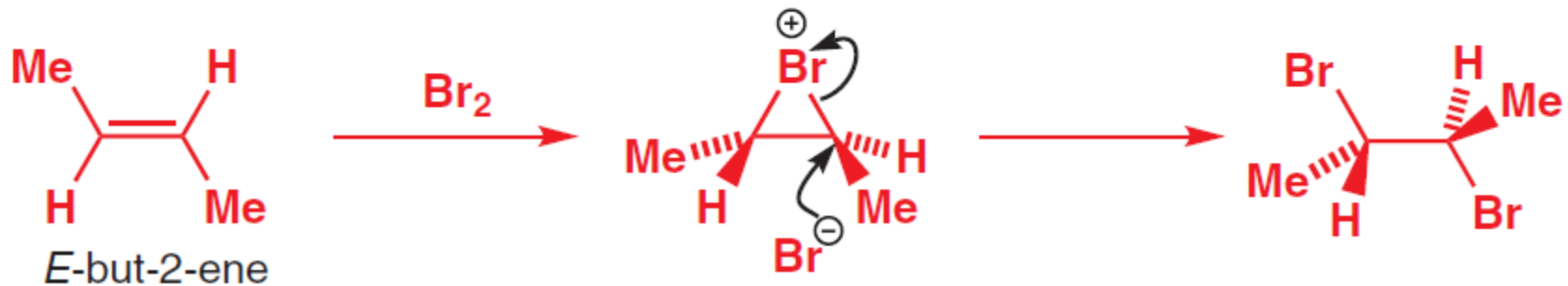
Z型烯烃



对称面

C2对称

E型烯烃



C2对称

对称面



二溴反式加成的立体化学画法

drawing the product like this shows clearly that there is overall *anti* addition of Br₂ across the *Z* double bond.

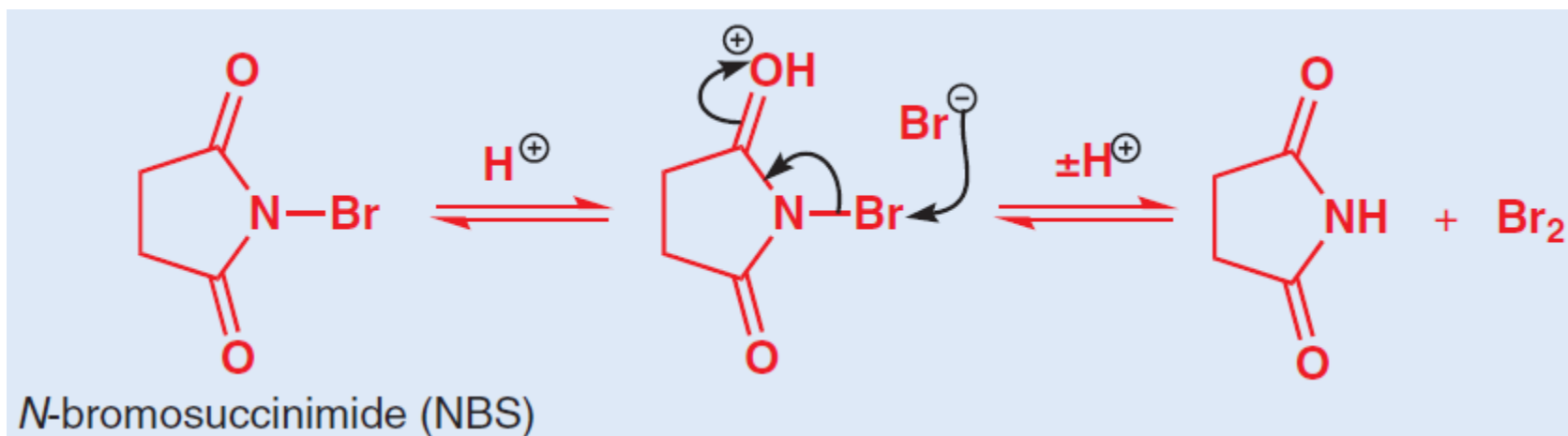
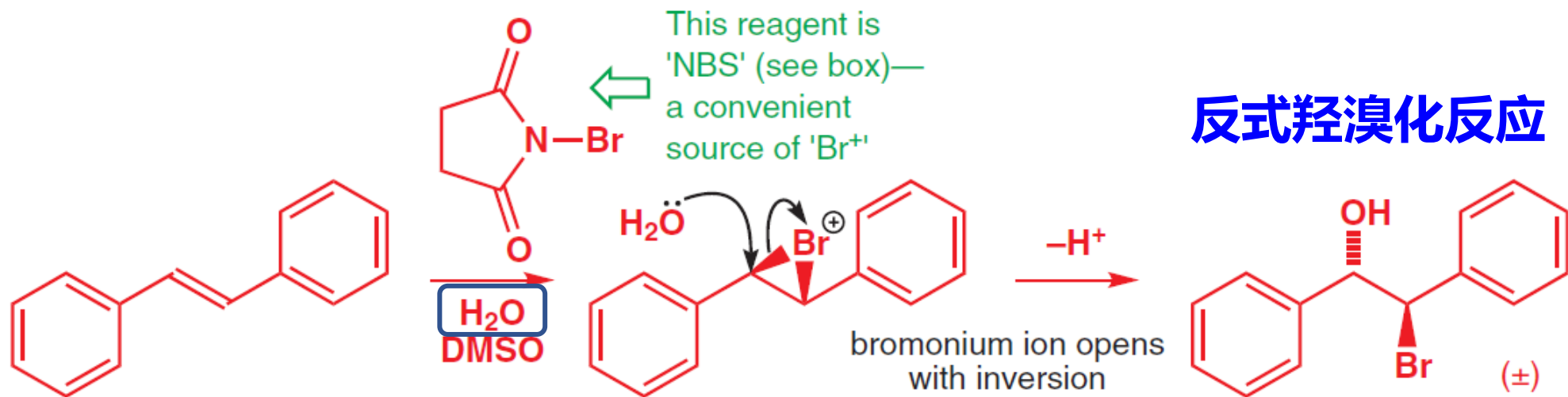


drawing the product like this shows clearly that there is overall *anti* addition of Br₂ across the *E* double bond.

this diastereoisomer is achiral (a *meso* compound): dotted line shows plane of symmetry



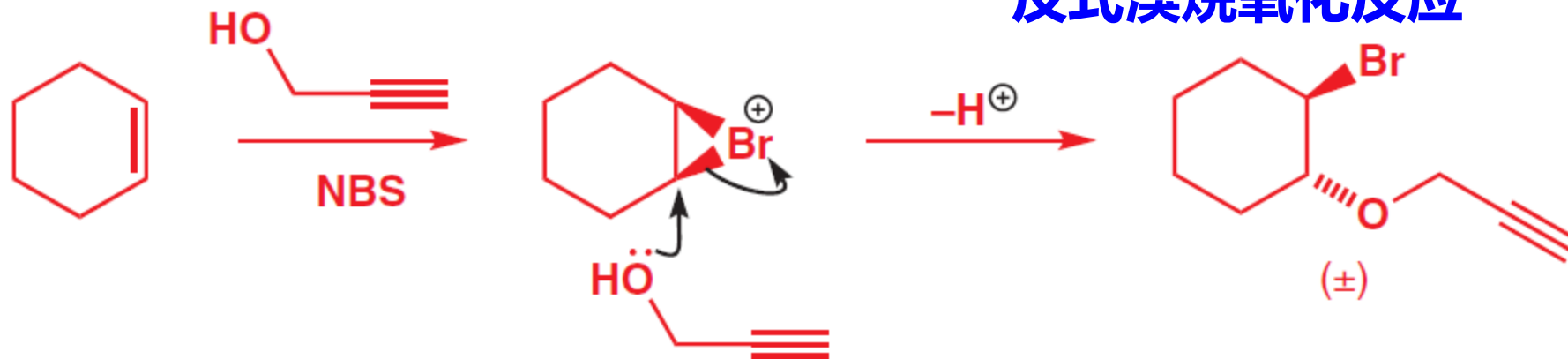
NBS作为溴正离子来源



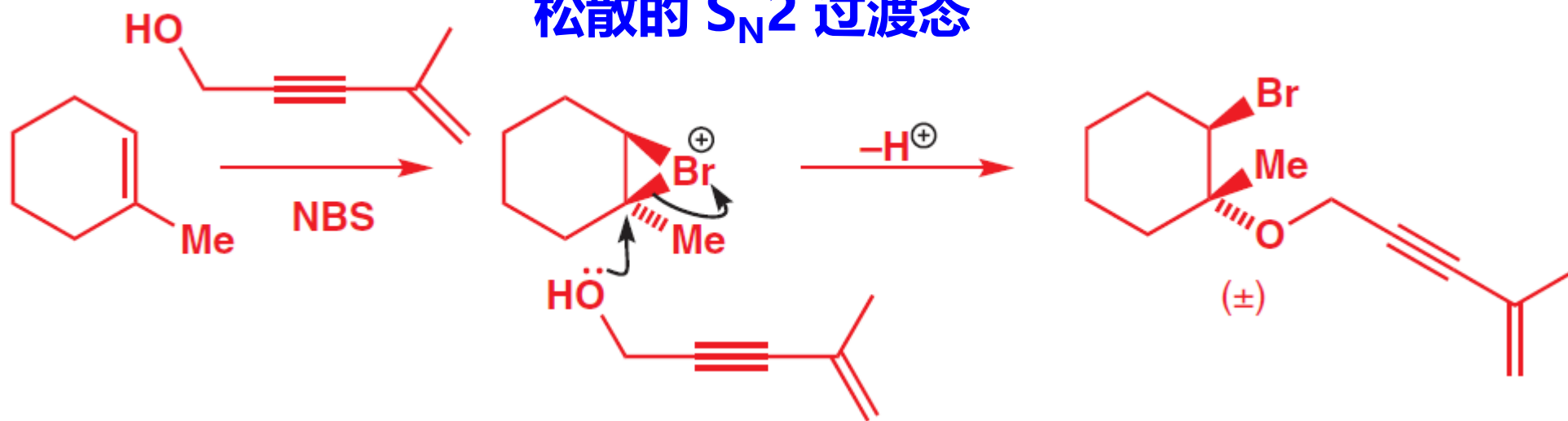


NBS作为溴正离子来源

反式溴烷氧化反应



松散的 S_N2 过渡态

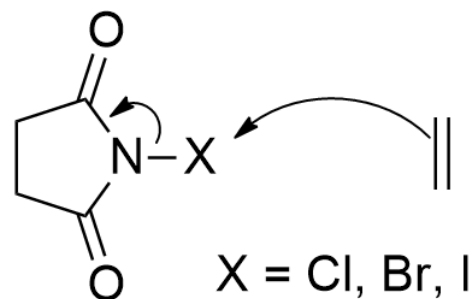


选择位阻较大一侧进攻溴鎓离子

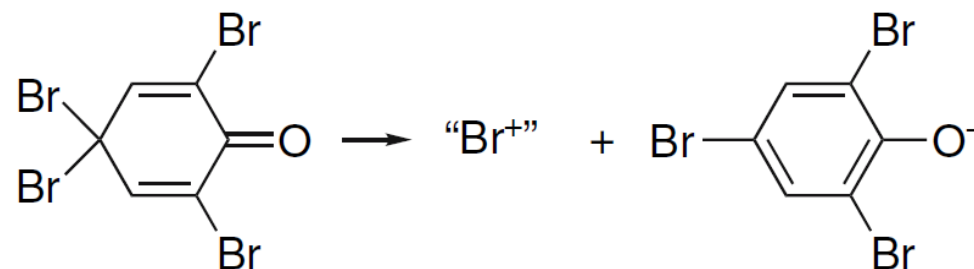


其他的卤正离子

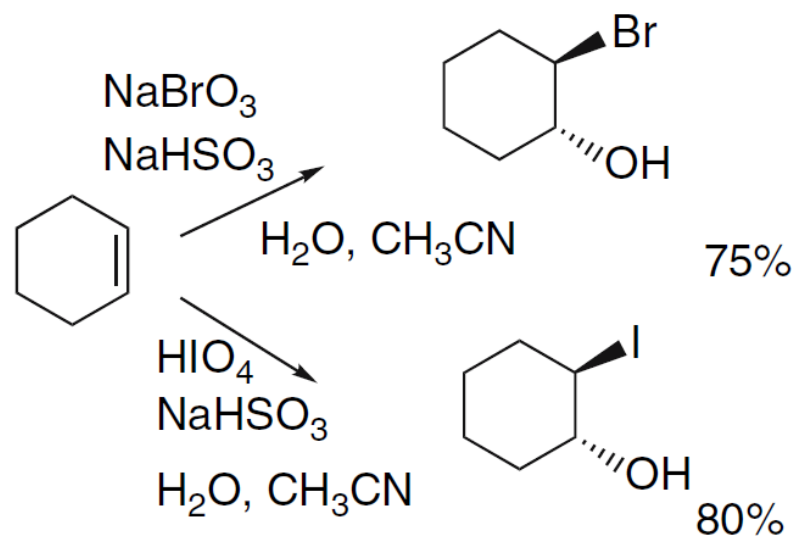
- N-卤代琥珀酰亚胺 (NBS, NCS, NIS)



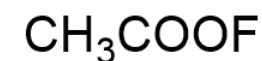
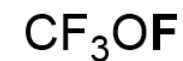
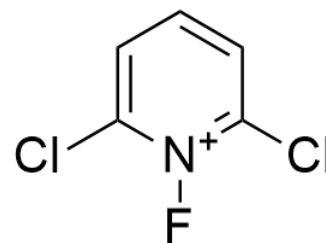
- 2,4,4,6-tetrabromocyclohexadienone



- 次卤酸 (HO-X)

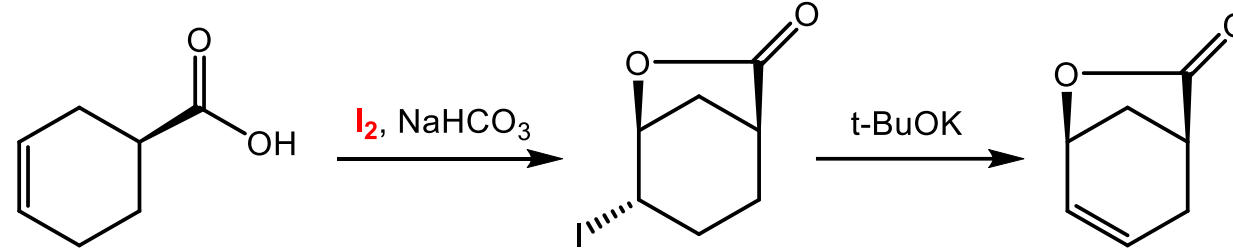


- "F⁺" agent

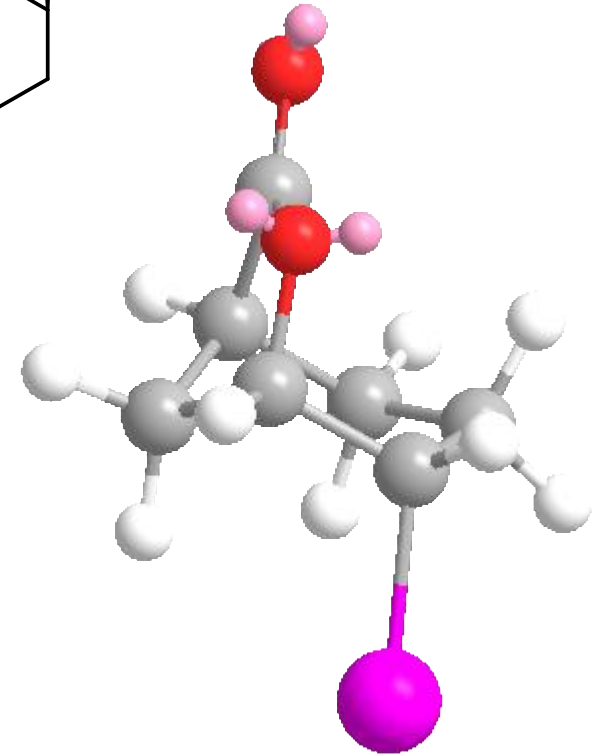
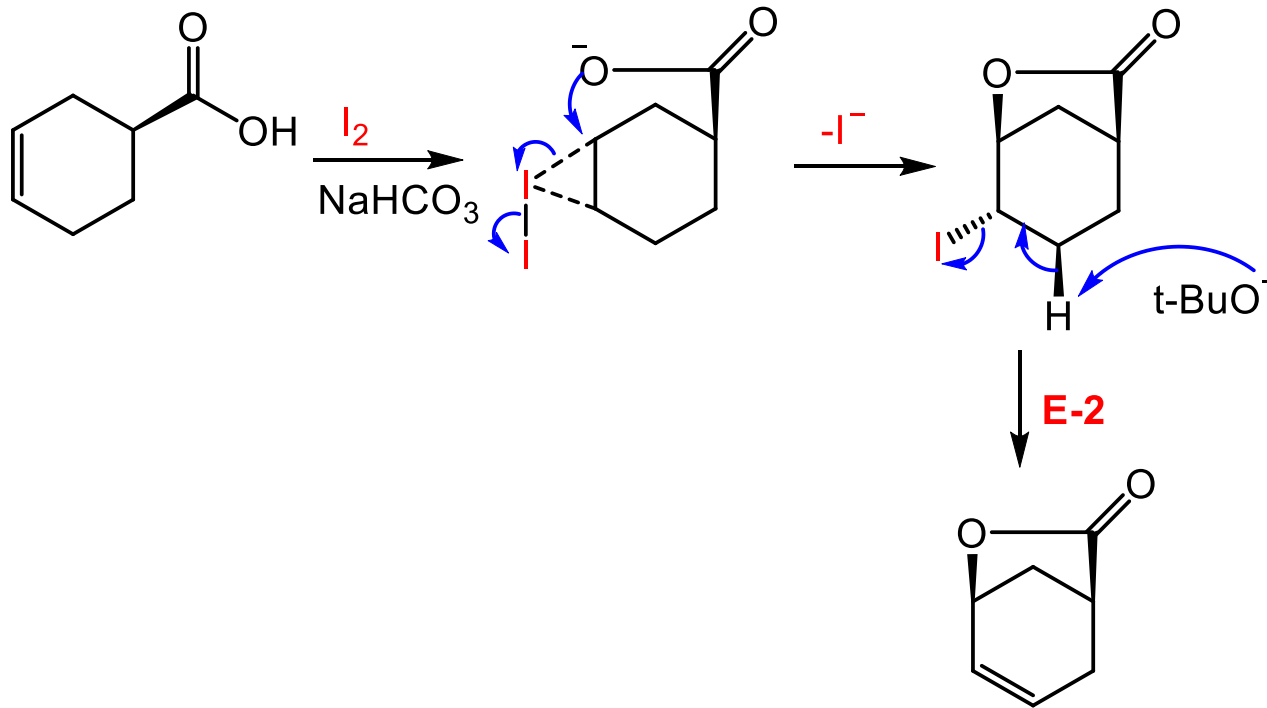




练习：画出反应机理



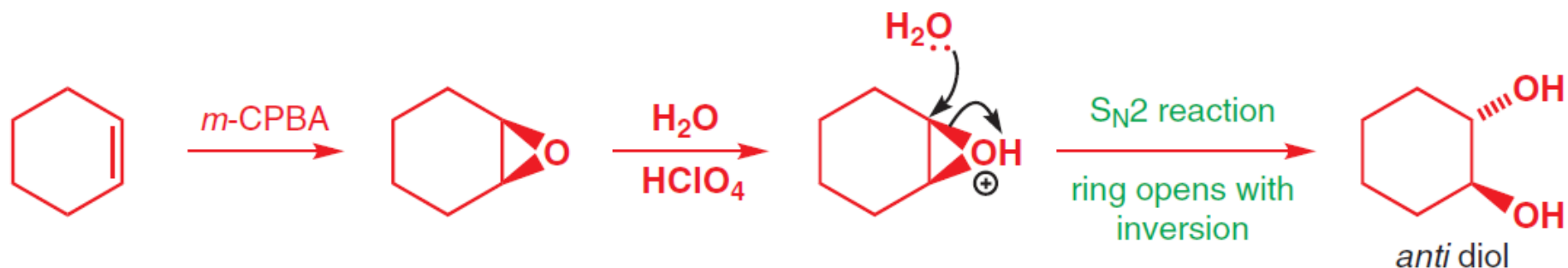
Answer:



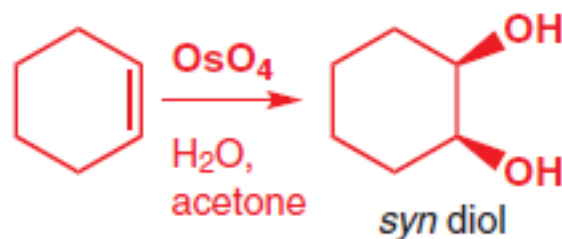


6.7 烯烃的双羟化反应

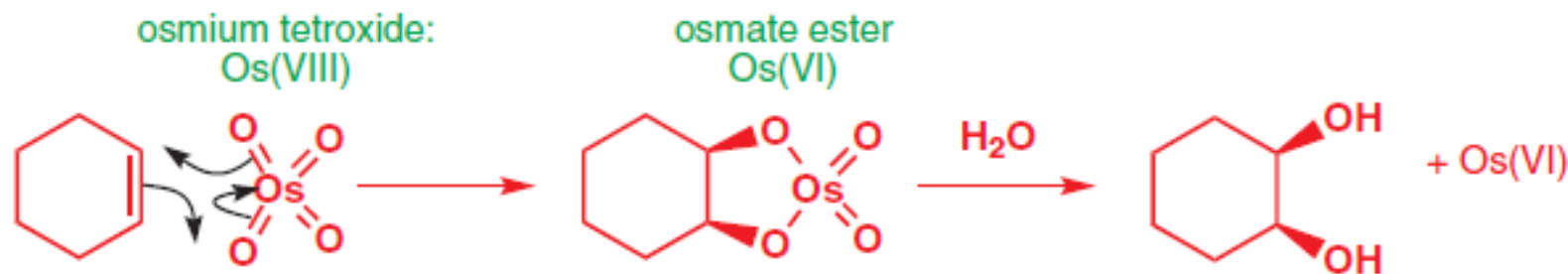
- 环氧化后，加水开环得到反式邻二醇



- OsO_4 氧化得到的是顺式邻二醇



OsO_4 剧毒、易挥发

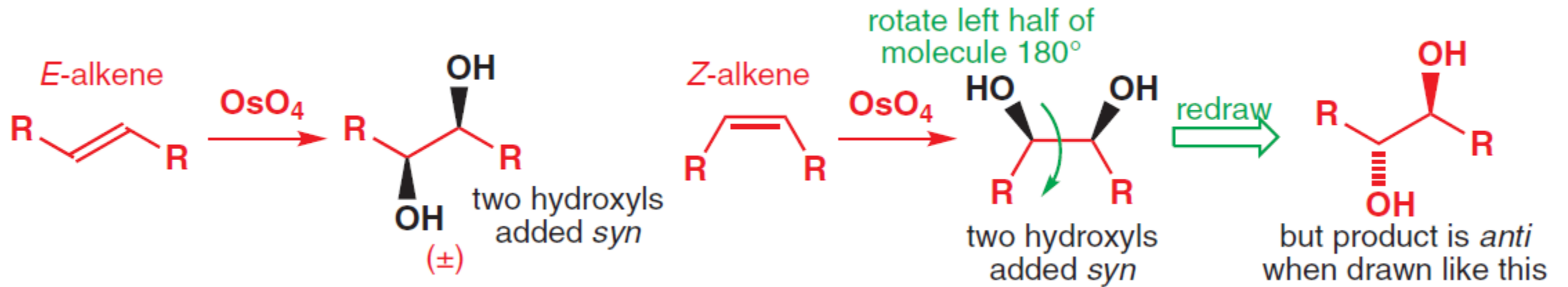


协同
顺式
环加成

其他方法：稀冷 KMnO_4



双羟化反应的立体化学

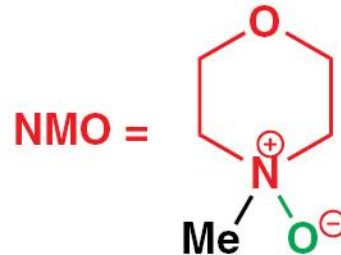
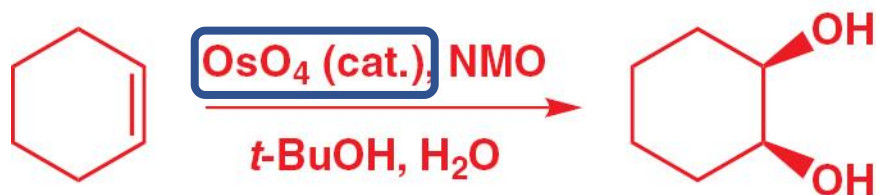


协同、顺式、环加成、构型保持



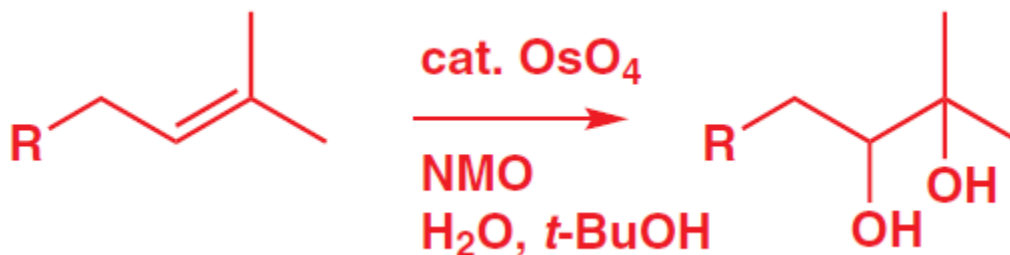
催化量 OsO_4

- 由于 OsO_4 **剧毒**、且昂贵，在反应过程中通常采用**加入共氧化剂**，将 OsO_4 的用量减少到催化量。
- 这些氧化剂可以是**双氧水**、**NMO**(氮氧化物)、过氧叔丁醇(**$t\text{BuOOH}$**)、铁氰化钾(**$\text{K}_3\text{Fe}(\text{CN})_6$**)等。



N-methyl
morpholine-
N-oxide

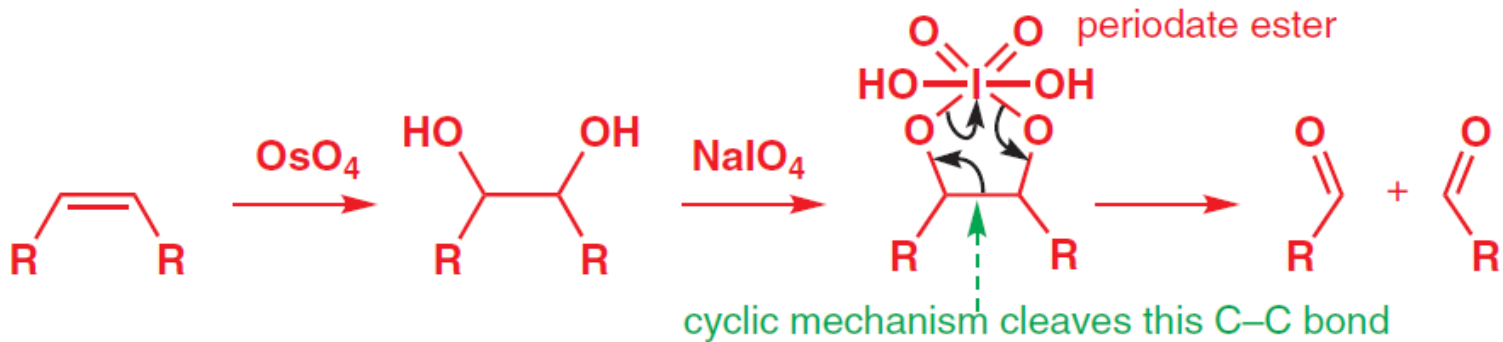
共氧化剂



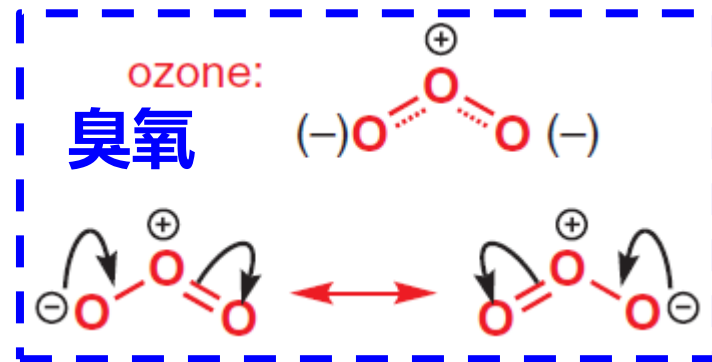
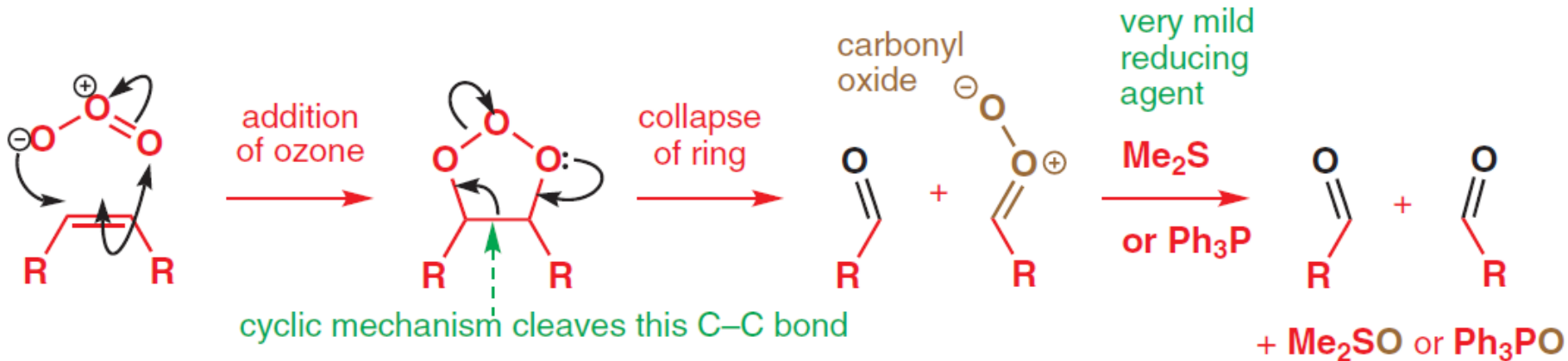


6.8 双键的切断反应: 高碘酸 (钠) 和臭氧化

策略一: 双羟化+高碘酸 (钠) 切断



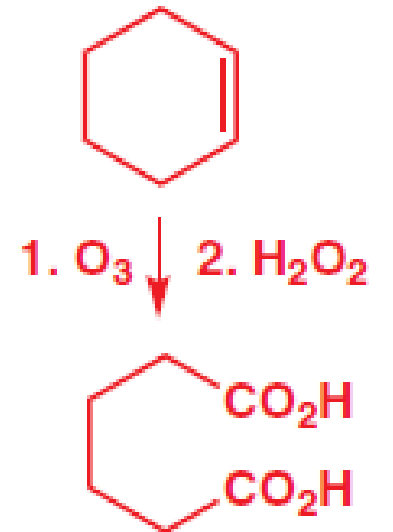
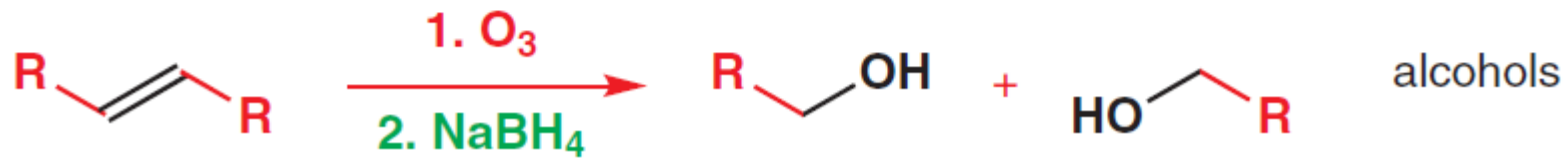
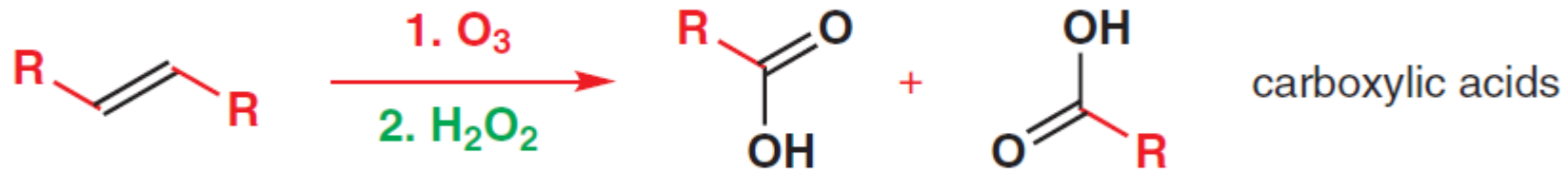
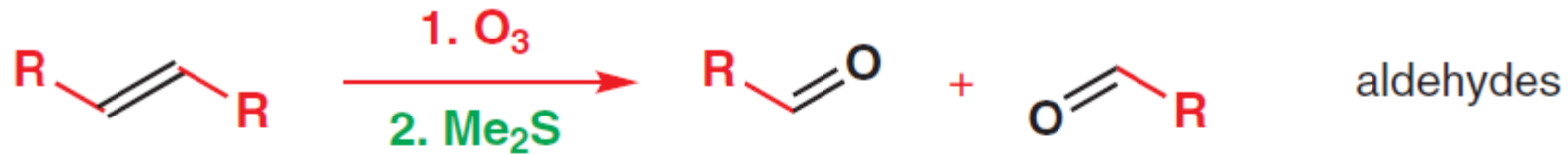
策略二: 臭氧化+还原





臭氧化反应不同的后处理产生不同产物

- 1) 温和还原剂锌粉、二甲硫醚、三价磷化合物、 Na_2SO_3 等：形成羰基化合物
- 2) 强的还原剂如 NaBH_4 ：形成醇。
- 3) 氧化剂如过量的双氧水：形成羧酸

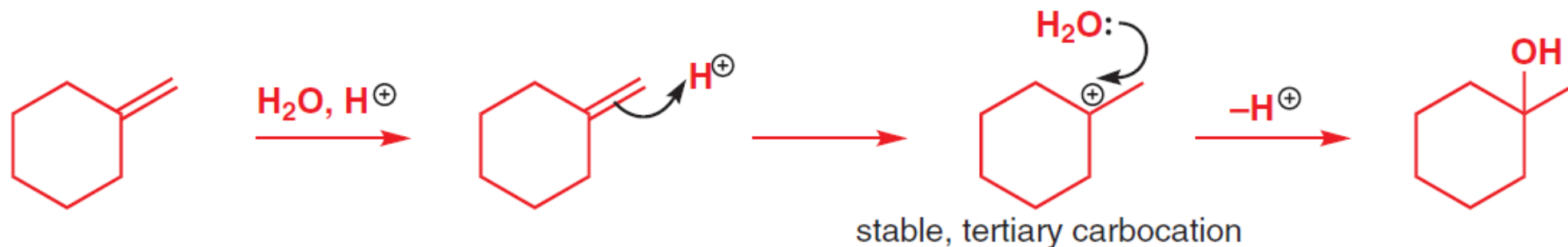




6.9 烯烃的水合反应

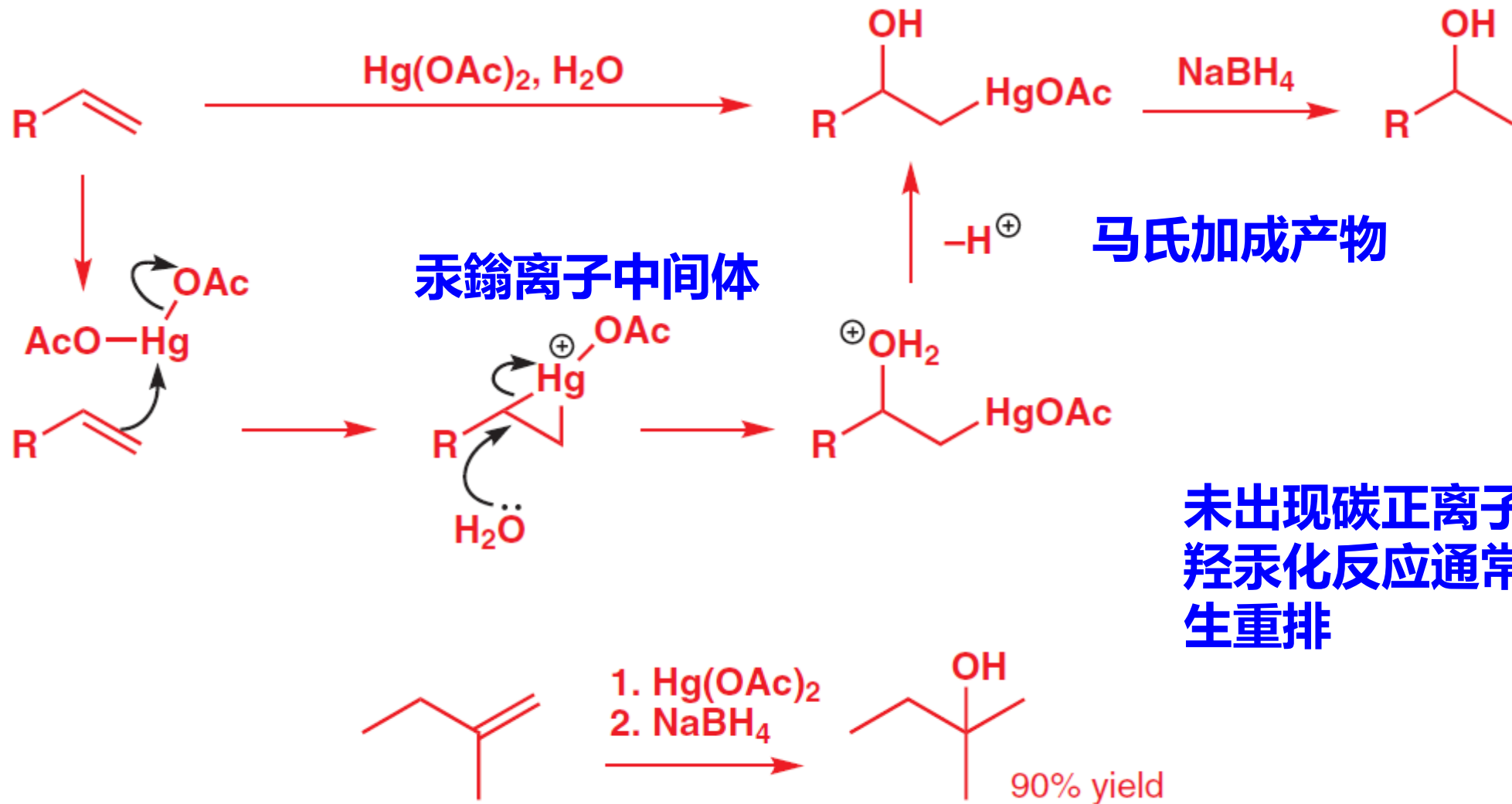


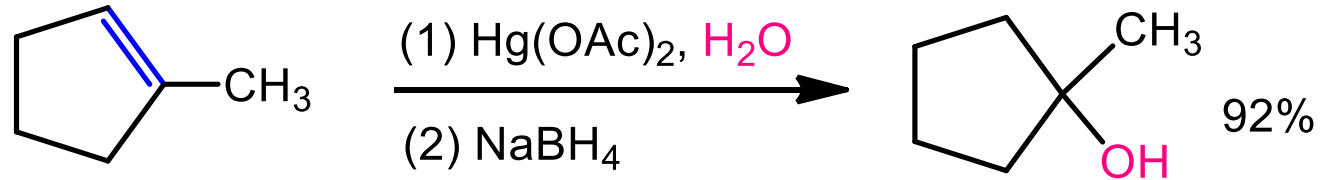
通常而言，很难判断在酸的水溶液条件下，烯烃是发生水合产生醇，还是醇脱水产生烯烃。



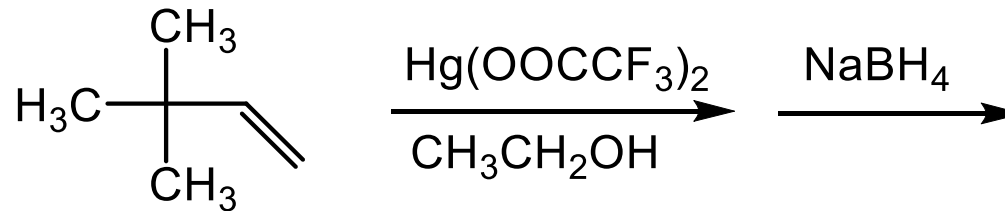


氧汞化-还原反应





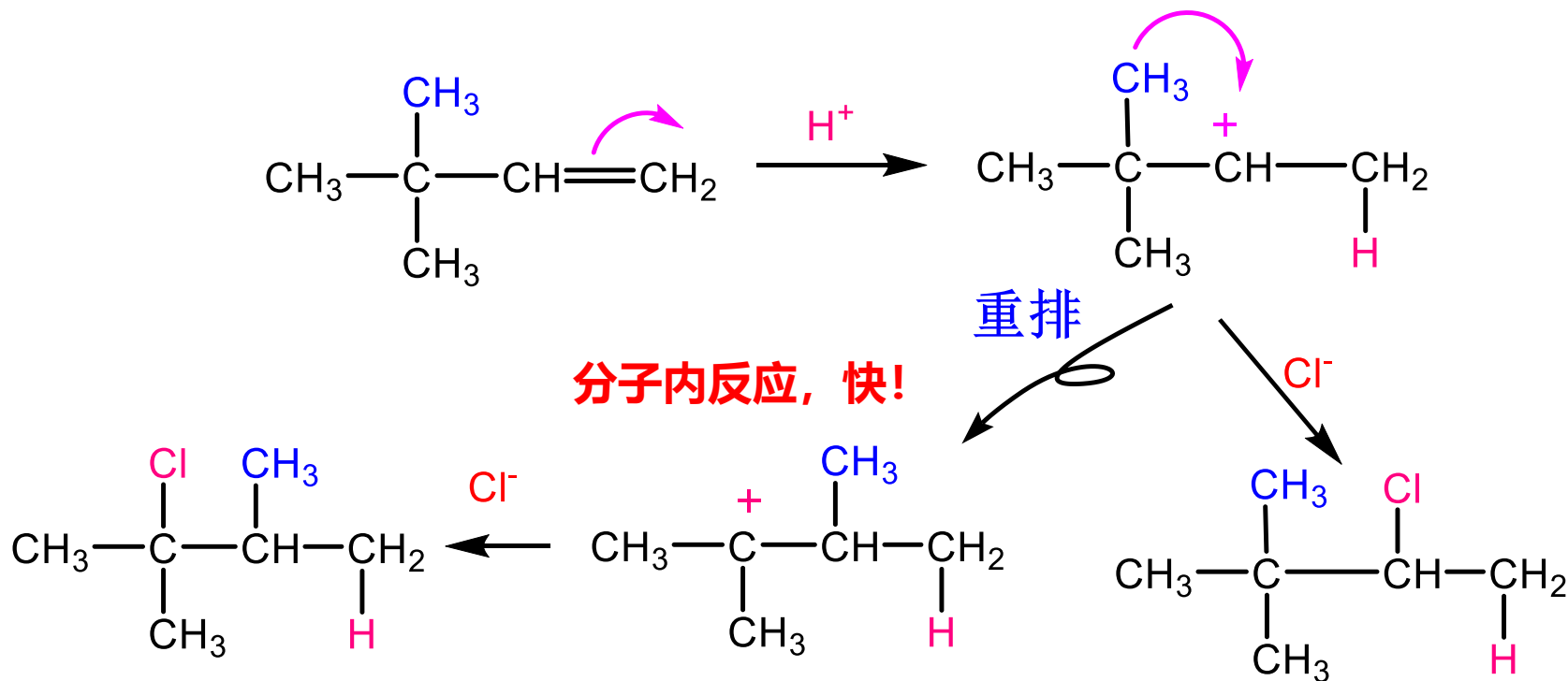
Question: 给出产物画出机理



If Hg salt replaced with H_2SO_4 , H_2O ?



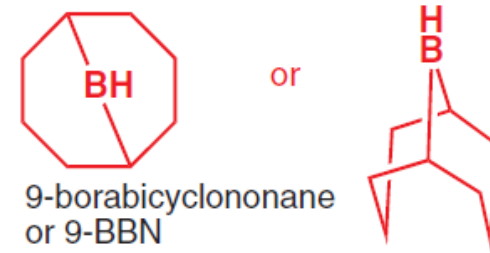
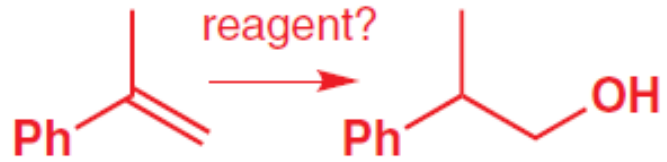
重排反应



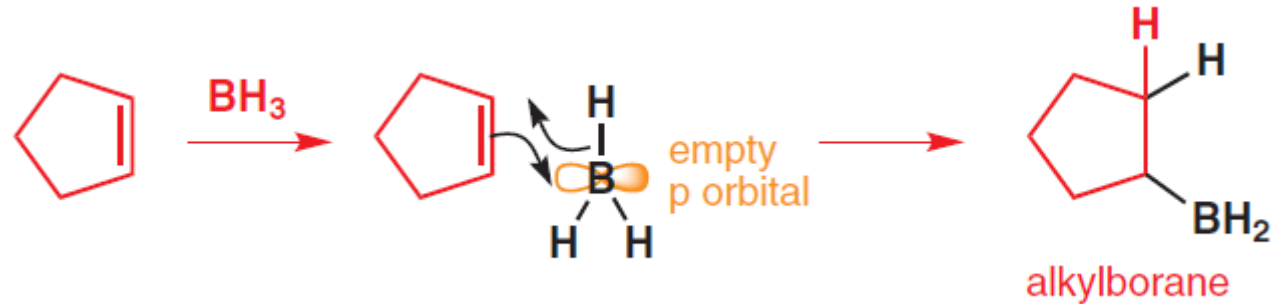
- **重排反应** —— 在化学键的断裂和形成过程中，组成分子的原子配置方式发生了改变，从而形成组成相同，结构不同的新分子。



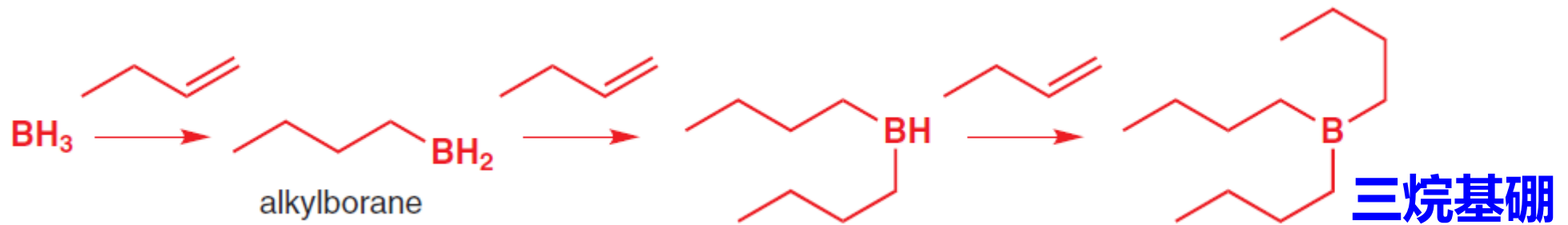
硼氢化反应



如何实现反马氏水合反应?



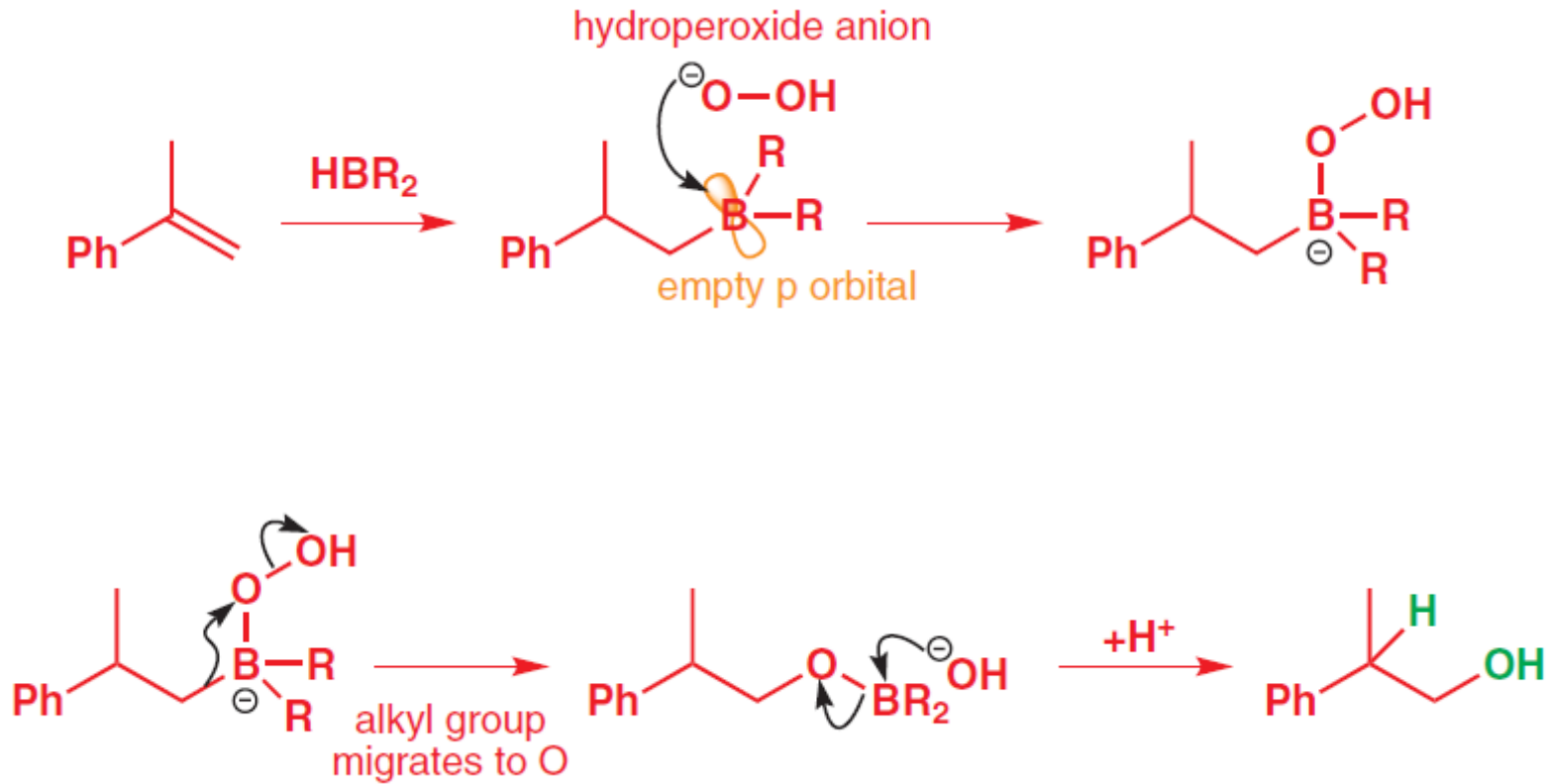
协同的同面加成



三个硼氢键都可使用



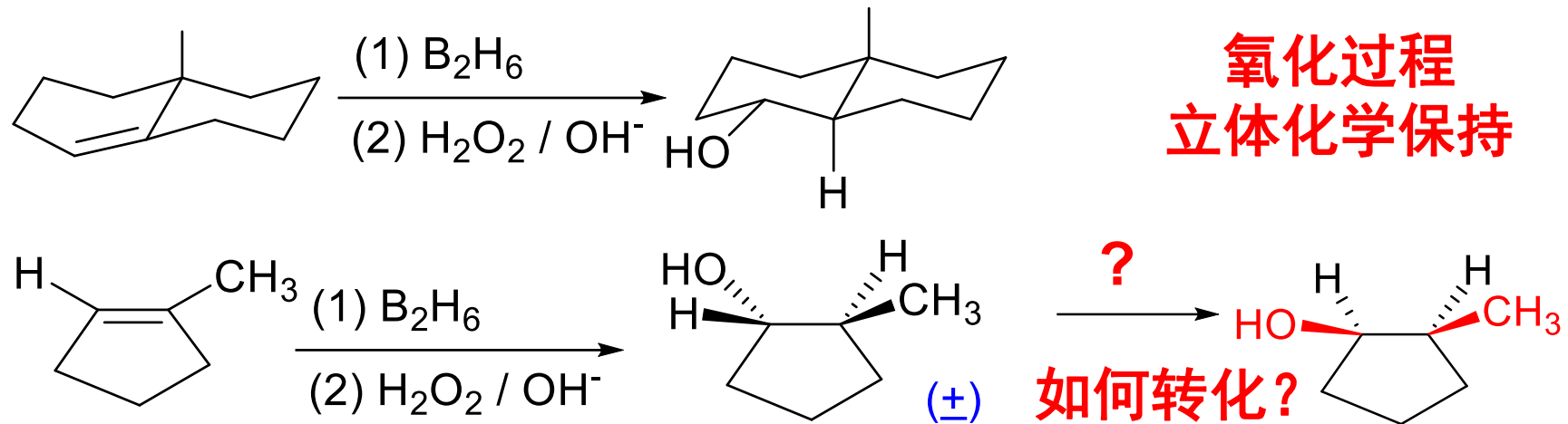
硼氢化-氧化制备伯醇



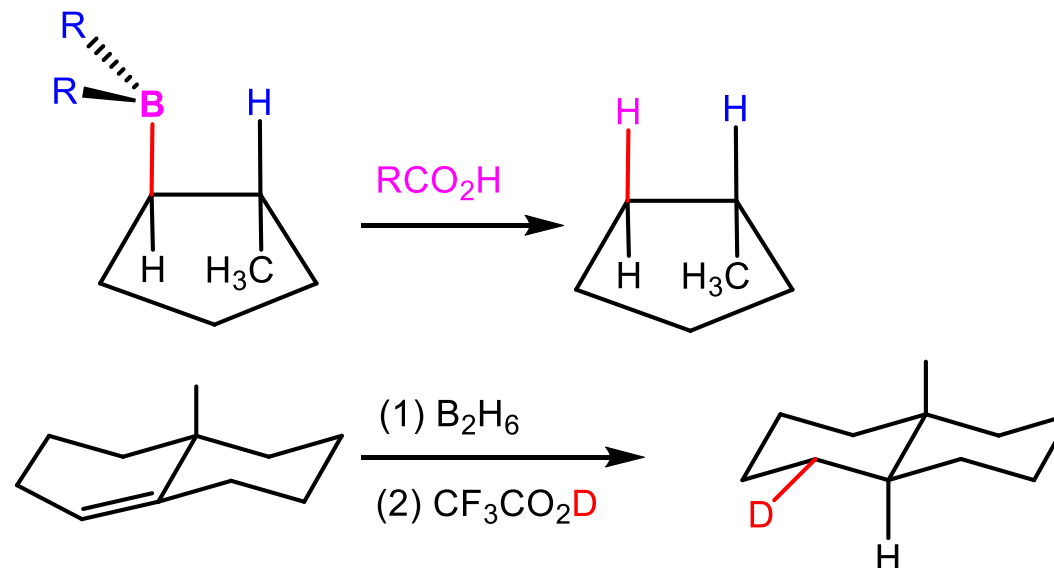
直接水解碳硼键得什么产物？



硼氢化-氧化的位置选择性和立体选择性



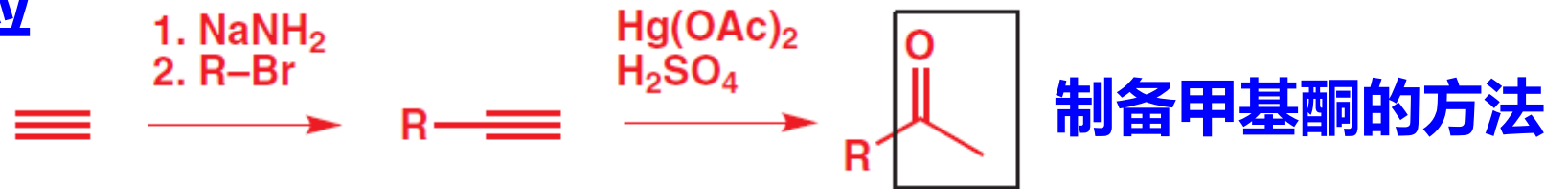
硼氢化-质子化:
实现烯烃还原



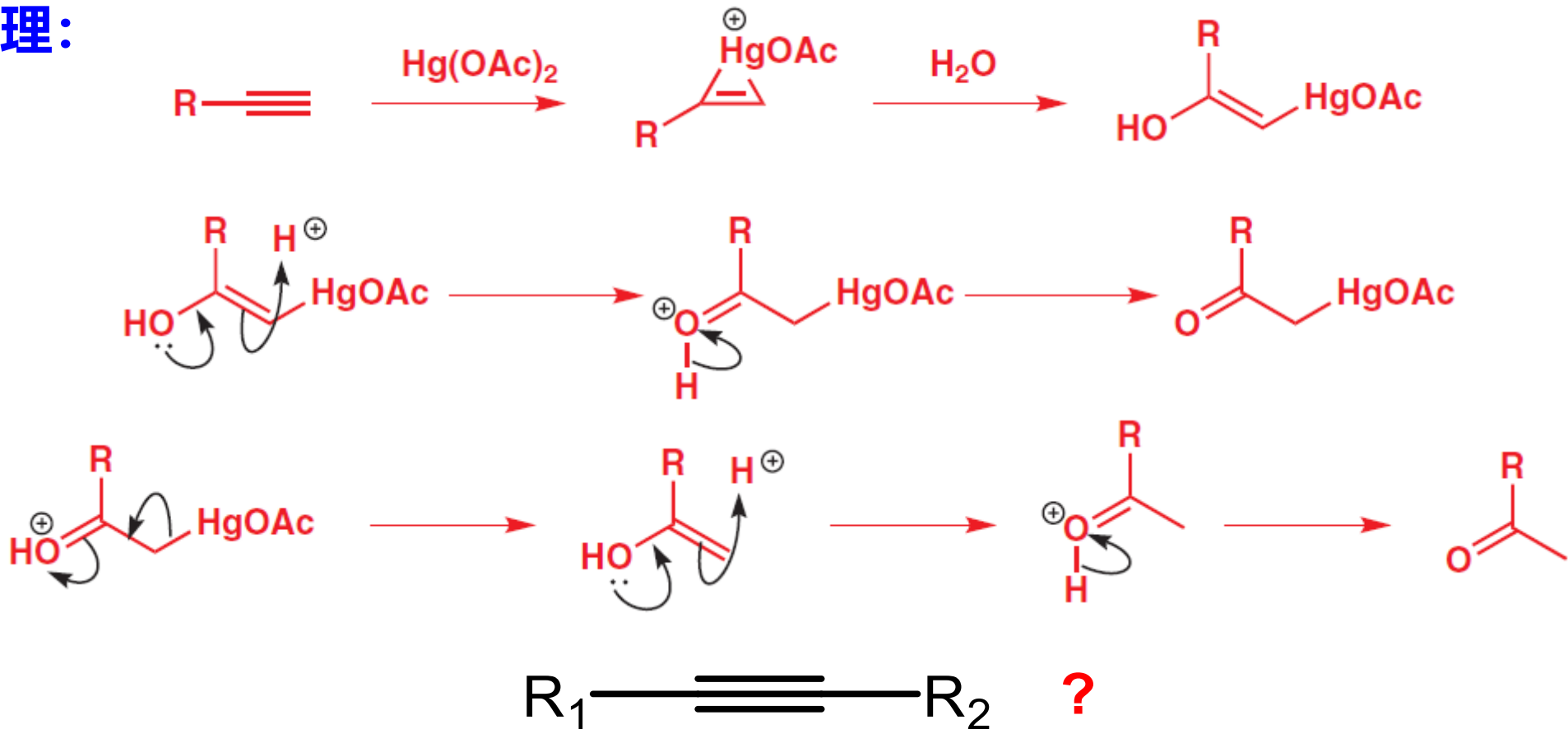


6.10 炔烃的亲电加成反应

炔烃的羟汞化-水解反应

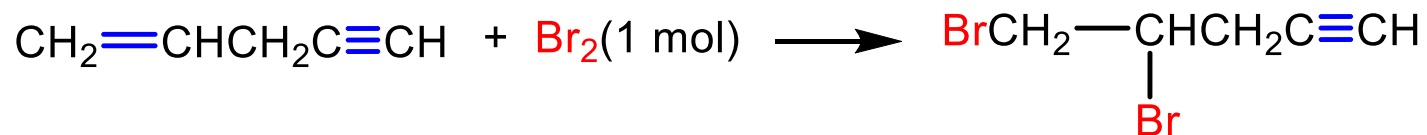
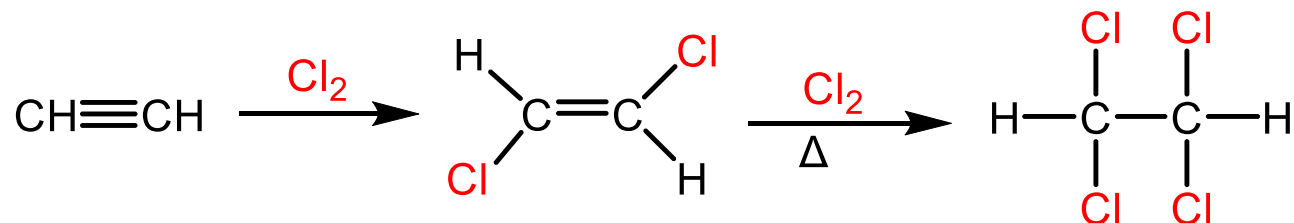
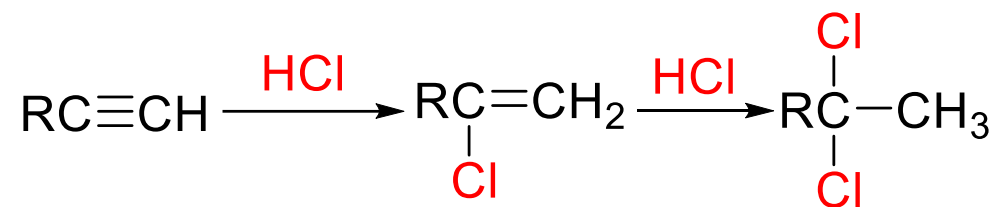


反应机理:

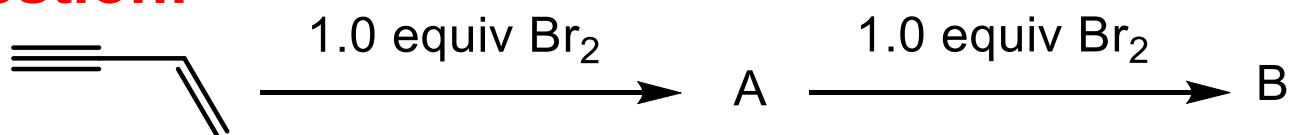




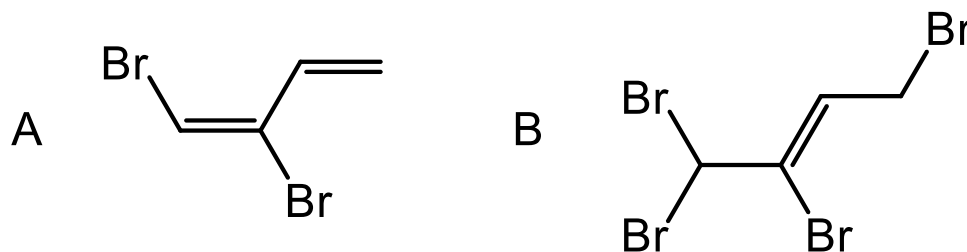
炔烃和HX、X₂的亲电加成反应



Question:

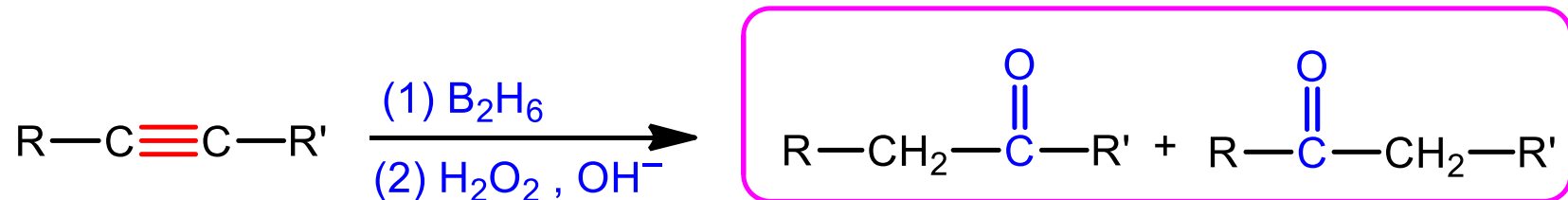
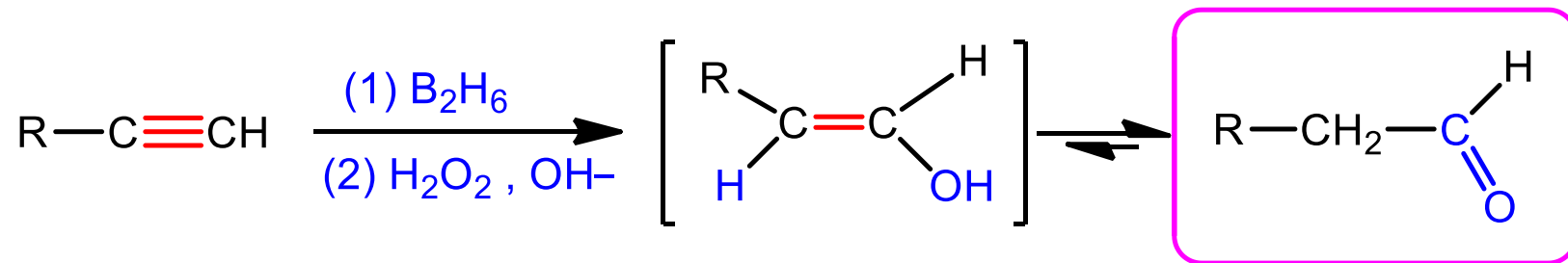
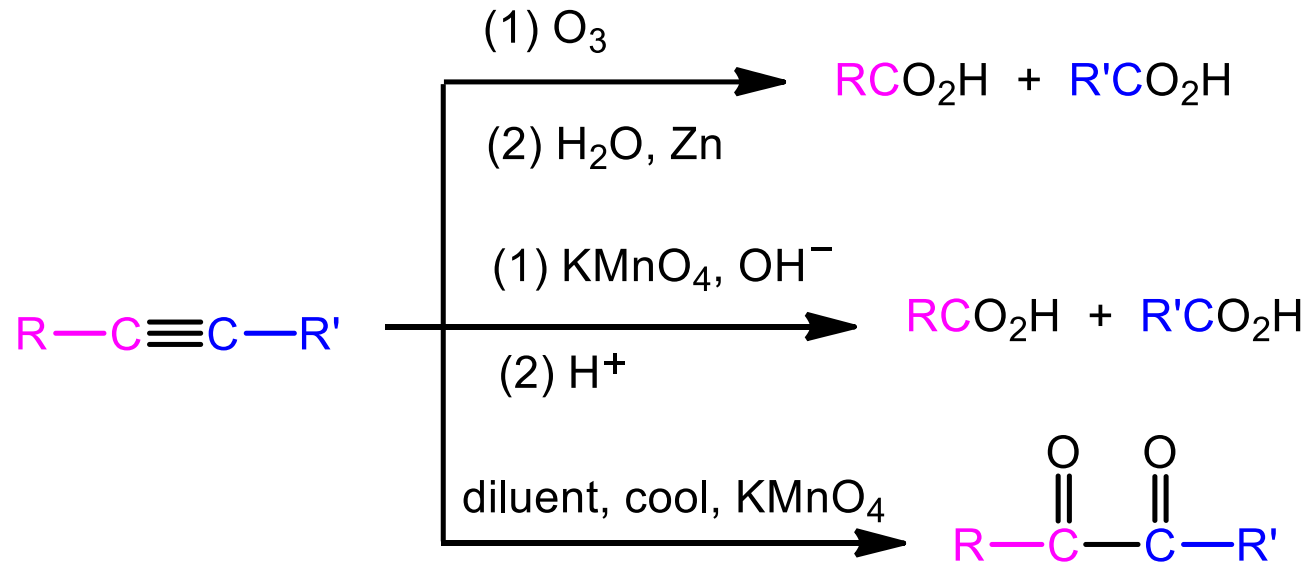


Answers:



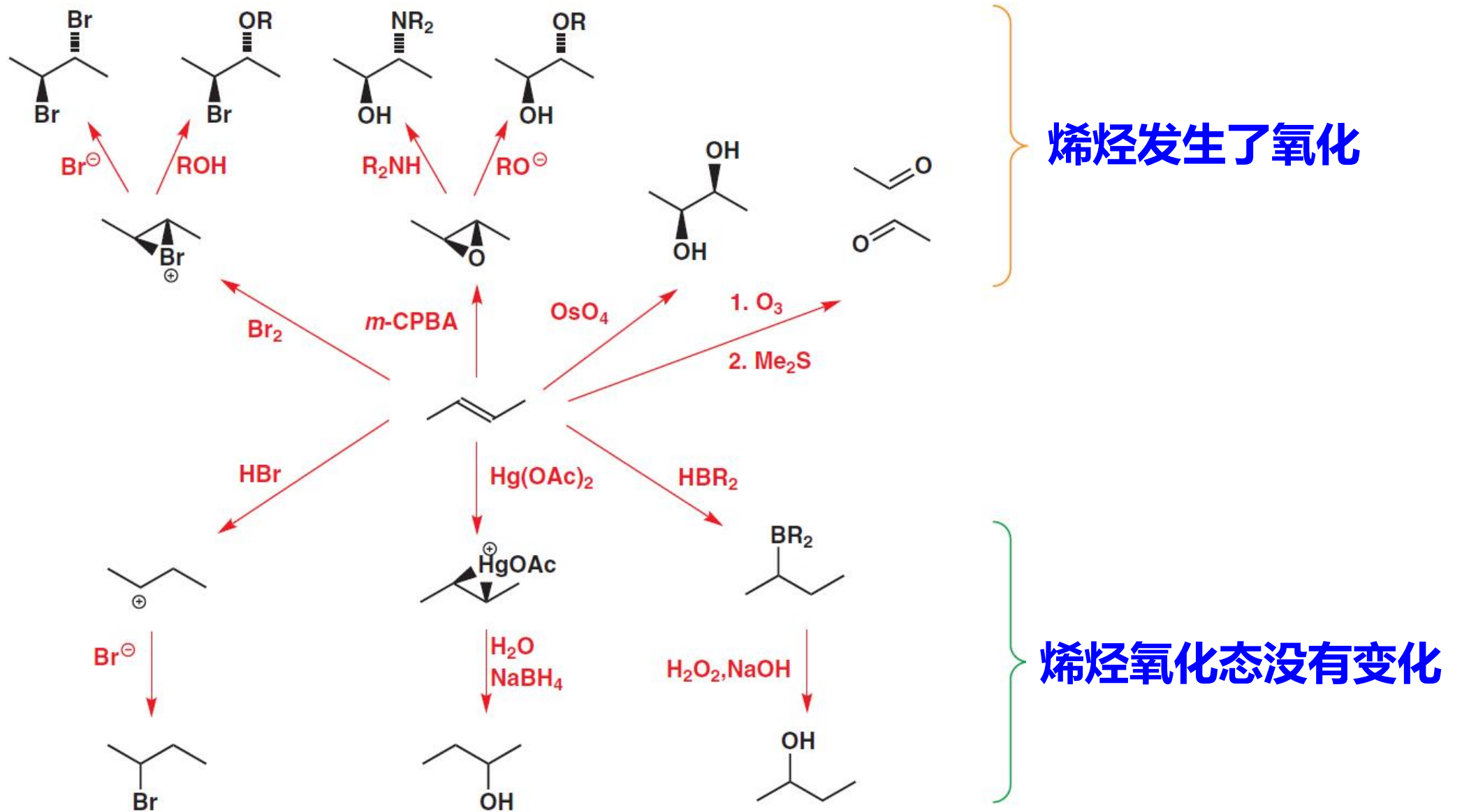


炔烃的氧化及硼氢化-氧化





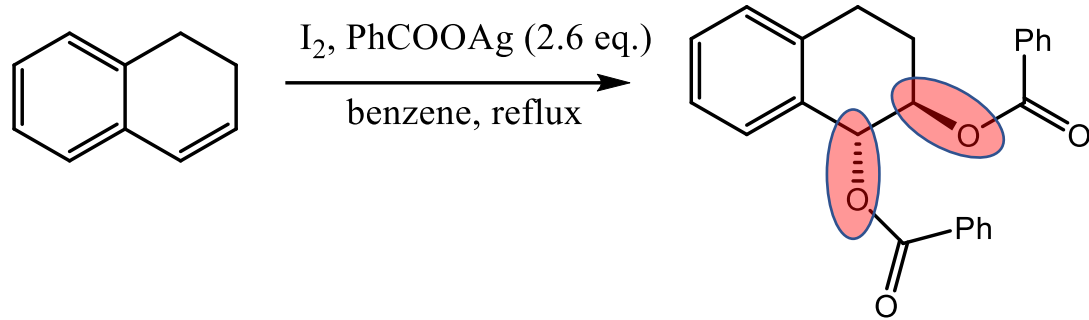
6.10 总结



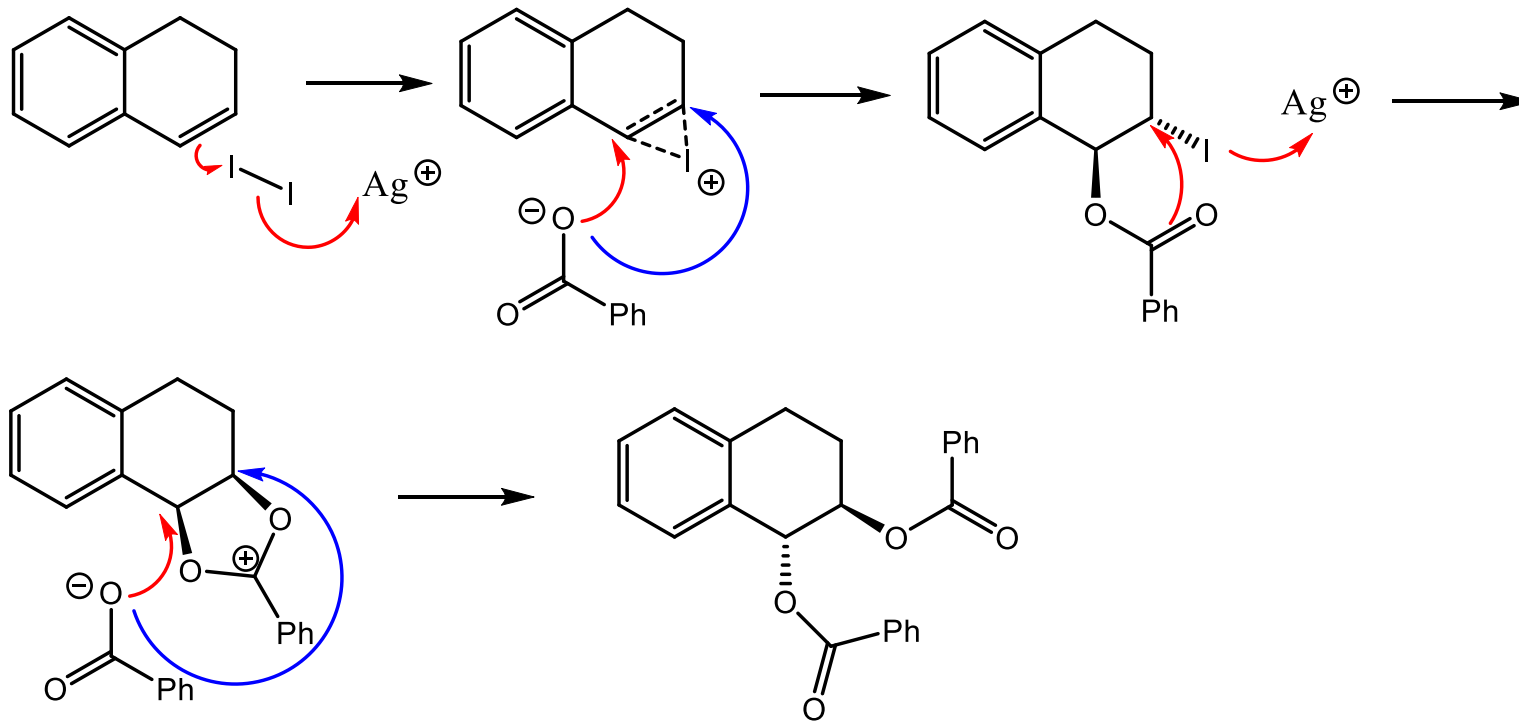


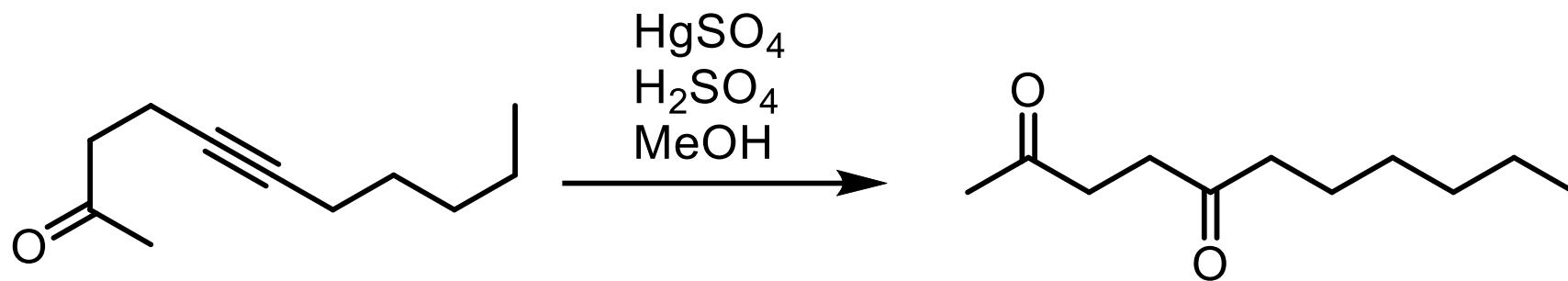
反应机理练习题

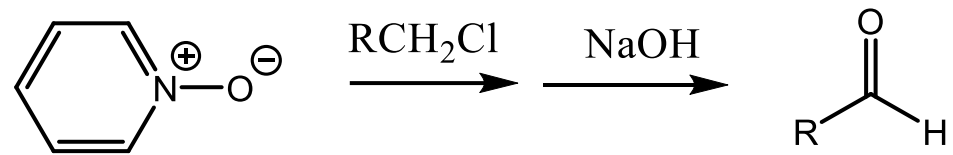
Prevost reaction



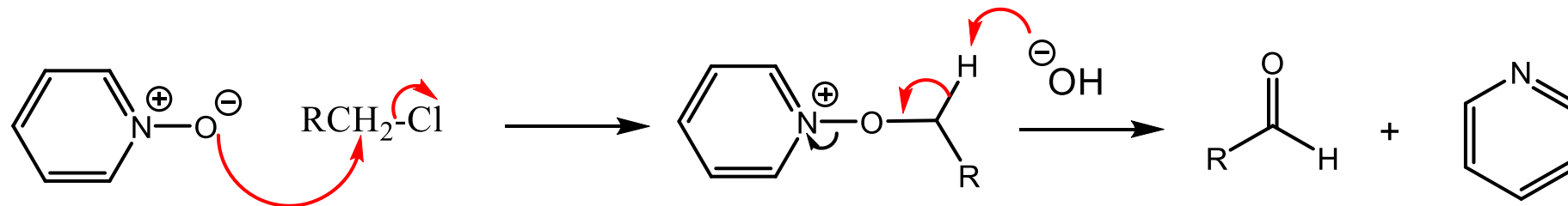
Mechanism:

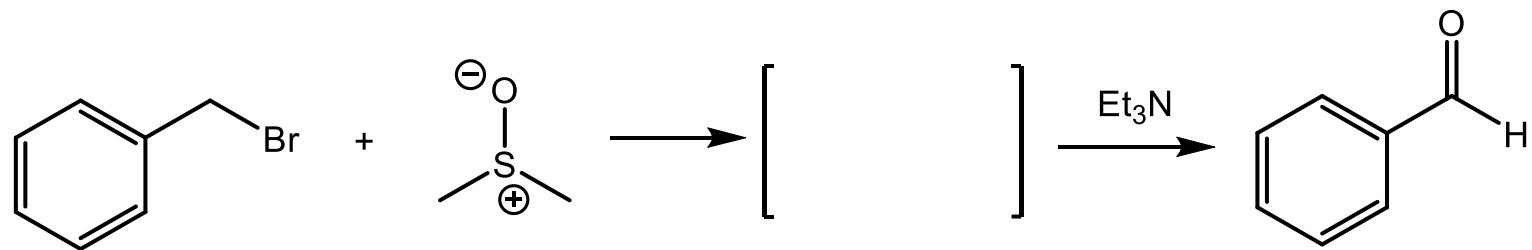






Mechanism:

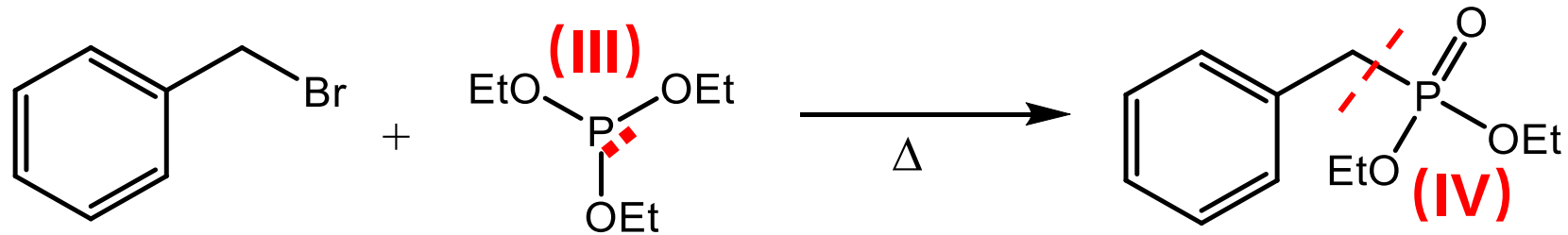




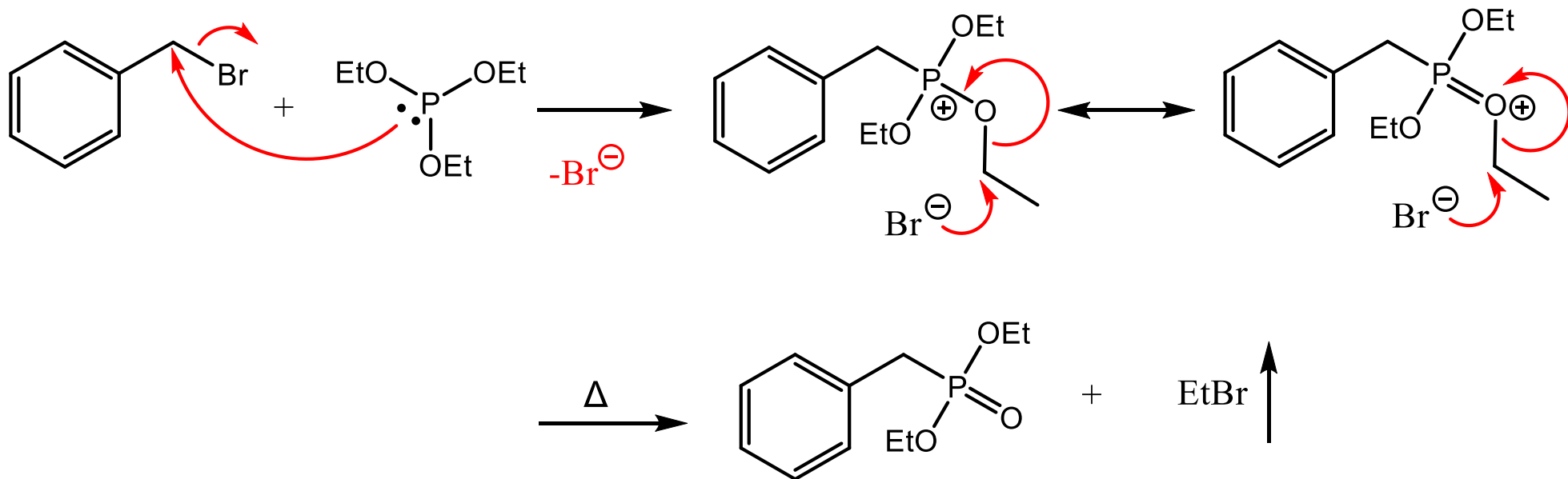
Mechanism:

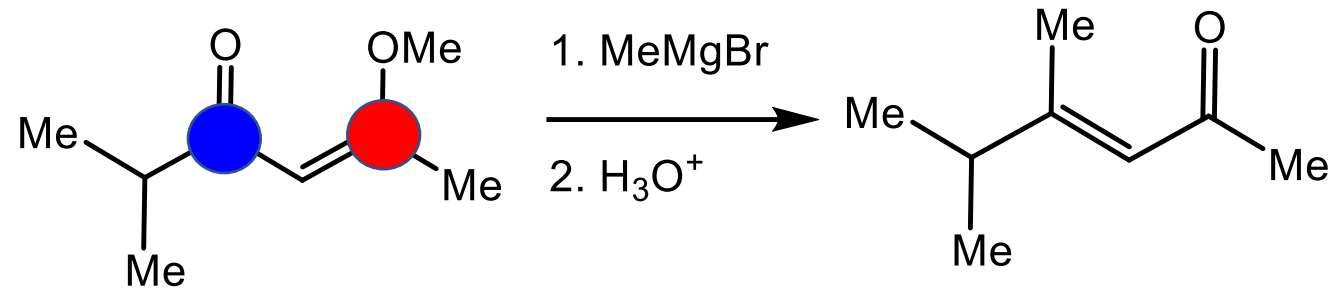


Abuzov reaction

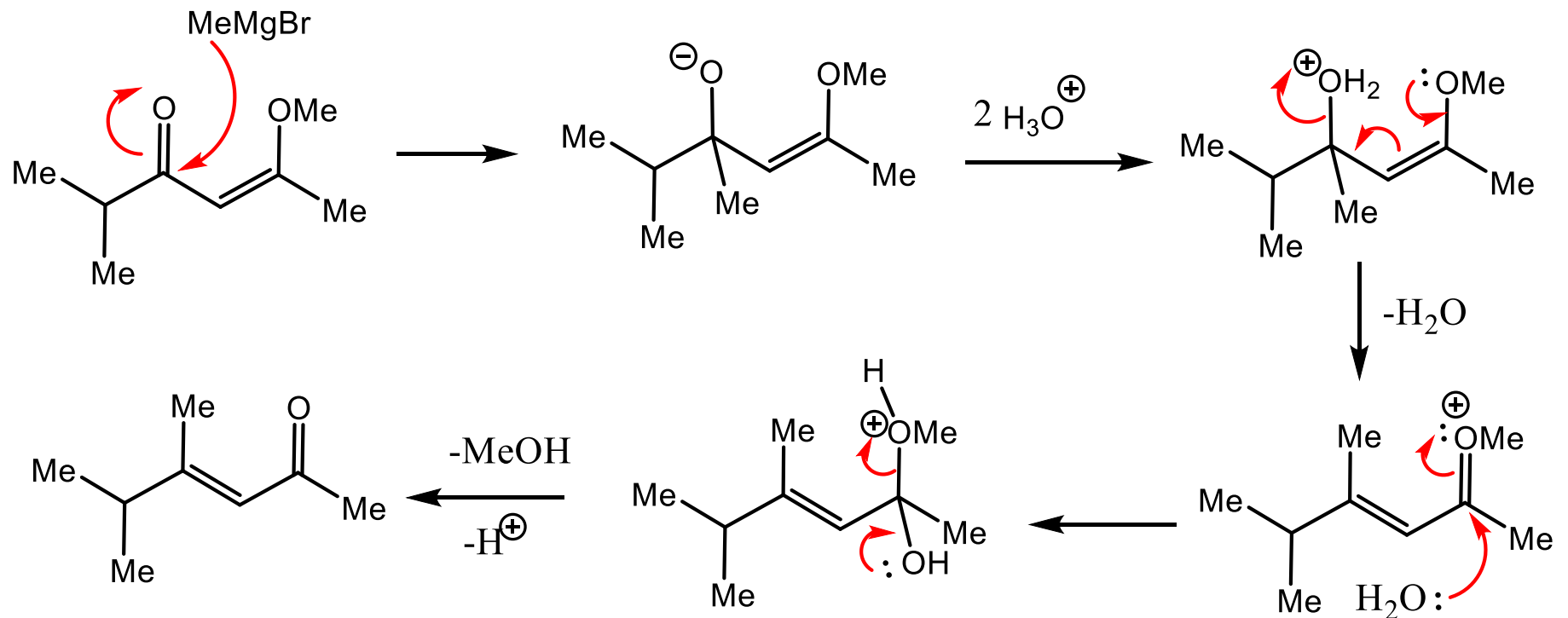


Mechanism:



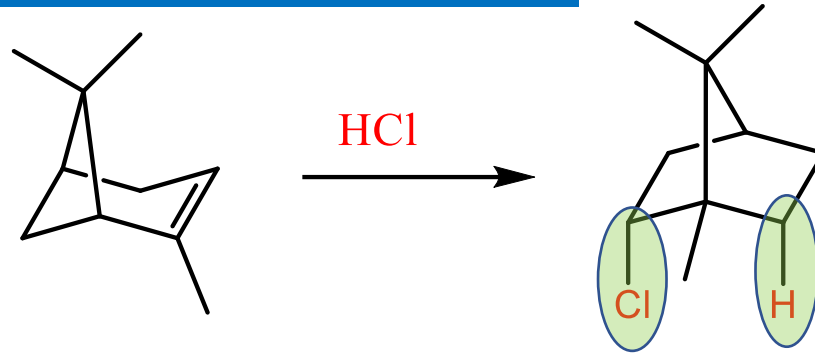


Mechanism:

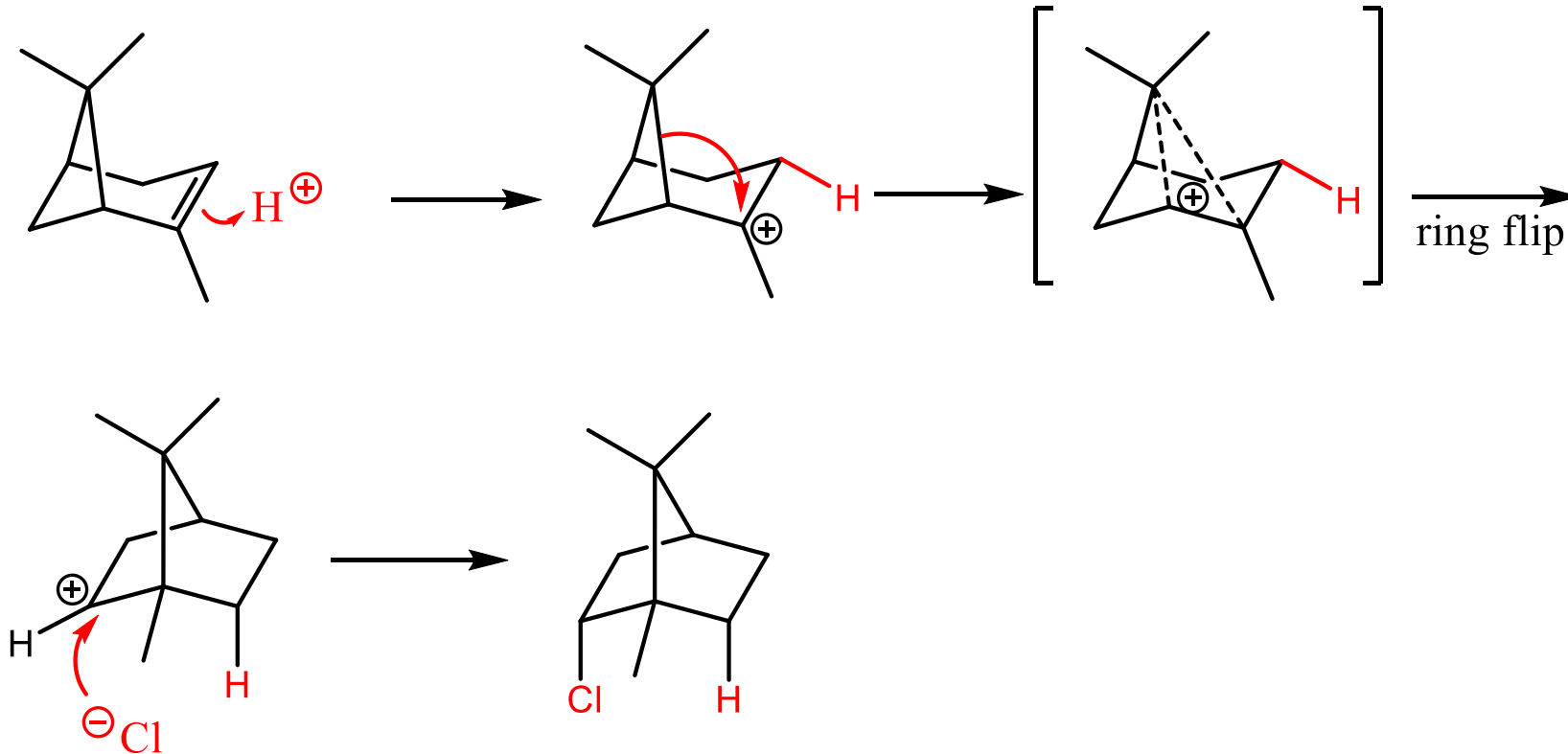




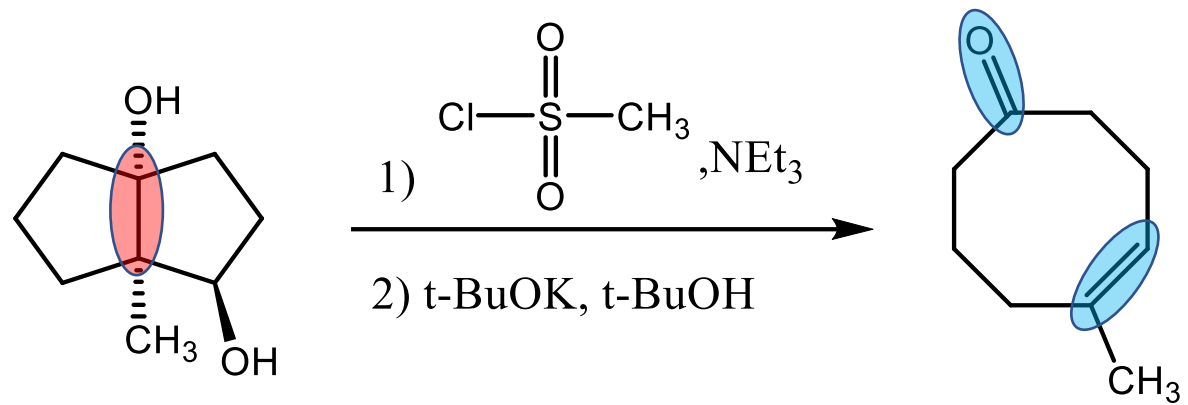
Wagner-Meerwein rearrangement



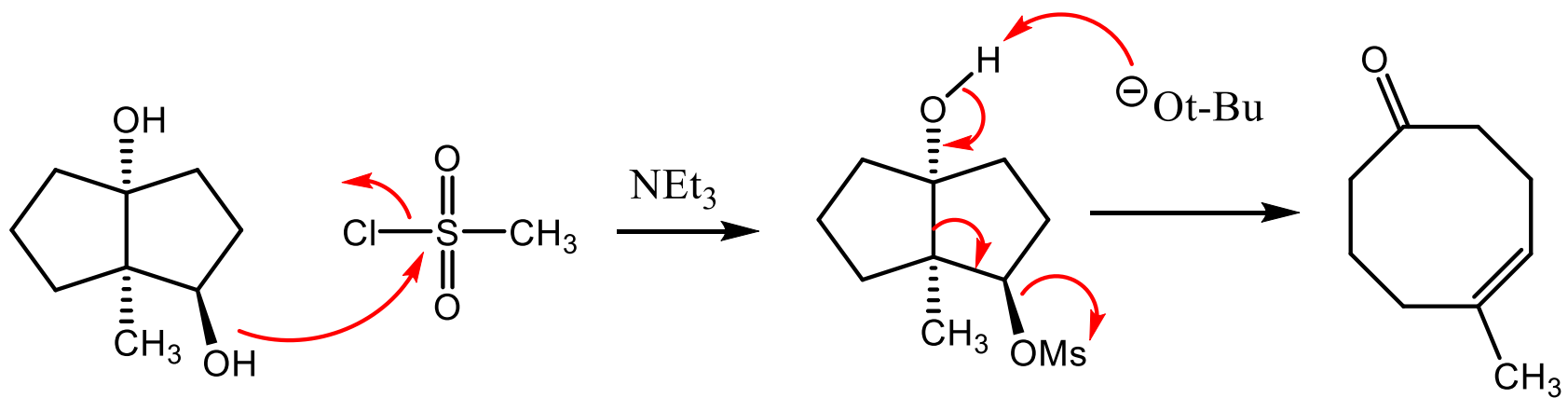
Mechanism:



Grob Fragmentation



Mechanism:





设计路线由 $\leq$ 三碳原料合成下列化合物

